PEARSON EDEXCEL INTERNATIONAL A LEVEL

PURE MATHEMATICS 1

EXPERTISE CLASSIFIED

Student Book

Edexcel paper progression • Through May/June 2026

THE MATHEMATICAL ROUTE

T — F — V — Q

94	813	12	74
questions	marks	topics	variants

EXPERTISE

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COURSE ROUTE

Contents

Expertise Classified organised by Topic, Family and Variant.

Membership follows the printed Edexcel paper progression: Question 6 through the end of each paper. Topic 09 is listed transparently with zero current Expertise questions.

Each question appears once in the course book. Cross-topic and additional Variant demands are shown on its route page.

T01

Indices and Surds

Questions drawn from printed Question 6 onward, following the paper's progression.

1	7	1	3
questions	marks	families	variants

T01

1 questions and 7 marks occur in the Expertise positions for this topic. 3 of 7 observed Variants are represented.

COVERAGE MAP

Topic 01 Expertise Coverage

Indices and Surds

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Index Laws and Re-expression

V01 Algebraic Fractional and Negative Powers	not selected in this set
V02 Re-expression Through a Named Power	not selected in this set
V03 Single Common-base Index Equation	not selected in this set

F02 • Power-substitution Equations

V01 Quadratic in an Exponential Power	not selected in this set
V02 Fractional-power Equation to a Polynomial	not selected in this set

F03 • Exact Surd Algebra

V01 Simplify, Combine and Expand Surds	1 selected
V02 Rationalise a Binomial Surd Denominator	1 selected
V03 Solve an Exact Linear Surd Equation	not selected in this set
V04 Nested-surd Reciprocal Identity	1 selected

F03

Exact Surd Algebra

Simplify, rationalise and solve exact expressions involving square roots while preserving an exact requested form.

VARIANT ROUTE

V04

Nested-surd Reciprocal Identity

1 selected question

1 Expertise questions 7 total marks

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(b) Express $\dfrac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

T02

Quadratics

Questions drawn from printed Question 6 onward, following the paper's progression.

6 questions	46 marks	4 families	6 variants

T02

6 questions and 46 marks occur in the Expertise positions for this topic. 6 of 10 observed Variants are represented.

COVERAGE MAP

Topic 02 Expertise Coverage

Quadratics

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Quadratic Equation Methods

V01 Direct Exact Quadratic Solution	not selected in this set
V02 Square-root or Fractional-power Substitution	1 selected
V03 Even-power or Biquadratic Substitution	not selected in this set
V04 Odd-power Quadratic Substitution	1 selected

F02 • Completed-square Form and Parabola Features

V01 Complete the Square, Read and Sketch	not selected in this set
V02 Complete the Square, then Intersect and Define a Region	1 selected

F03 • Construct a Quadratic from Graph Data

V01 Roots, Symmetry and a Point	1 selected
V02 Vertex Form and a Point	1 selected

F04 • Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count	2 selected
V02 Curve Tangency or Separation	not selected in this set

F05 • Quadratic Modelling from Constraints

V01 Sum-product Geometric Constraints	not selected in this set

F01

Quadratic Equation Methods

Solve a quadratic directly or reveal one by substituting for a repeated expression, then map all admissible roots back to the original variable.

VARIANT ROUTE

V02

Square-root or Fractional-power Substitution

1 selected question

1 Expertise questions 5 total marks

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a, b and c are integers to be found.

(5)

F02

Completed-square Form and Parabola Features

Convert a quadratic to completed-square form, read its turning point and symmetry, and use these features in sketches, intersections or region descriptions.

VARIANT ROUTE

V02

Complete the Square, then Intersect and Define a Region

1 selected question

1 Expertise questions 10 total marks

FAMILY

11.

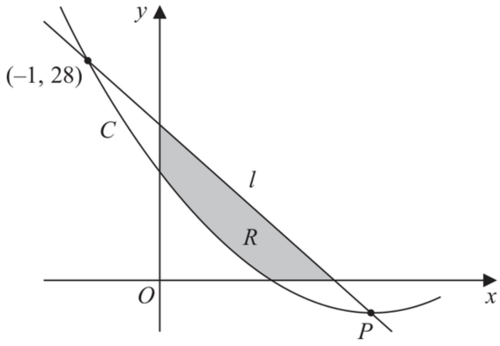


Figure 5

Figure 5 shows part of the curve C with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^2 - 12x + 14$$

(a) Write $2x^2 - 12x + 14$ in the form

$$a(x+b)^2 + c$$

where a, b and c are constants to be found.

(3)

Figure 5 shows part of the curve C with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^2 - 12x + 14$$

Given that C has a minimum at the point P

(b) state the coordinates of P

(1)

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

(c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

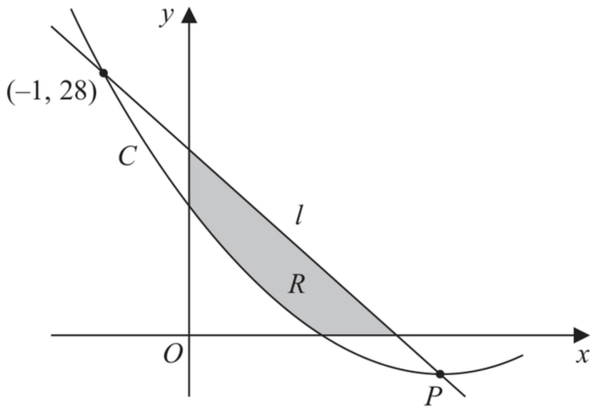


Figure 5

Figure 5 shows part of the curve C with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^2 - 12x + 14$$

The finite region R, shown shaded in Figure 5, is bounded by the x-axis, l, the y-axis, and C.

(d) Use inequalities to define the region R.

(3)

TOPIC 02 • FAMILY 03

F03

Construct a Quadratic from Graph Data

Select a root form or vertex form from graph information and determine its scale using another point or turning value.

VARIANT ROUTE

V01 **Roots, Symmetry and a Point** — 1 selected question

V02 **Vertex Form and a Point** — 1 selected question

2 Expertise questions — 18 total marks

FAMILY

8.

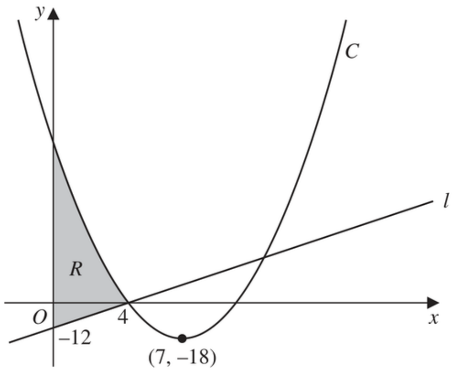


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C.

Given that l cuts the y-axis at –12 and cuts the x-axis at 4, as shown in Figure 2,

(a) find an equation for l, writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x-axis at 4 and at k, where k is a constant

(b) deduce the value of k,

(1)

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x-axis at 4 and at k, where k is a constant

(c) find $f(x)$.

(3)

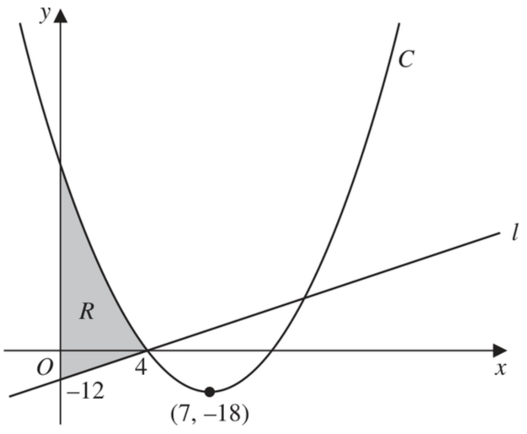


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C.

Given that l cuts the y-axis at -12 and cuts the x-axis at 4, as shown in Figure 2,

The region R is shown shaded in Figure 2.

(d) Use inequalities to define R.

(2)

Question 5

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at (2, 20)
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

9. The curve C_1 has equation $y = \mathrm{f}(x)$.

Given that

- $\mathrm{f}(x)$ is a quadratic expression
- C_1 has a maximum turning point at (2, 20)
- C_1 passes through the origin

(b) find an expression for $\mathrm{f}(x)$.

(3)

9. The curve C_1 has equation $y = \mathrm{f}(x)$.

Given that

- $\mathrm{f}(x)$ is a quadratic expression
- C_1 has a maximum turning point at (2, 20)
- C_1 passes through the origin

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

F04

Discriminant and Root-count Conditions

Translate a required number of real roots or curve intersections into a sign condition on the discriminant, then solve the resulting parameter equation or inequality.

VARIANT ROUTE

V01

Parameter Range from a Required Root Count

2 selected questions

2 Expertise questions 13 total marks

FAMILY

Question 6

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \qquad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0$$

(3)

$3p^2 - 10p - 8 > 0$

(b) Hence, using algebra, find the range of possible values of p

(3)

10. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \qquad \text{where } k \text{ is a constant}$$

(a) Find the exact solutions to the given equation for $k = 4.5$

(3)

10. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \qquad \text{where } k \text{ is a constant}$$

(b) Find the set of possible values of k for which the given equation has no real roots.

(4)

T03

Simultaneous Equations

Questions drawn from printed Question 6 onward, following the paper's progression.

2	14	1	2
questions	marks	families	variants

2 questions and 14 marks occur in the Expertise positions for this topic. 2 of 4 observed Variants are represented.

T03

COVERAGE MAP

Topic 03 Expertise Coverage

Simultaneous Equations

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Model-parameter Simultaneous Systems

V01 Two-observation Calibration and Prediction — not selected in this set

F02 • Nonlinear Systems by Substitution

V01 Linear plus Quadratic System — not selected in this set

V02 Polynomial Pair to a Biquadratic — 1 selected

V03 Rational-cubic Intersections with Follow-on Geometry — 1 selected

F02

Nonlinear Systems by Substitution

Isolate one variable or equate two curve expressions, reduce the system to a solvable quadratic structure, and back-substitute to obtain matched ordered pairs and any requested geometry.

VARIANT ROUTE

V02 **Polynomial Pair to a Biquadratic** — 1 selected question

V03 **Rational-cubic Intersections with Follow-on Geometry** — 1 selected question

2 Expertise questions — 14 total marks

FAMILY

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0$$

(2)

$$2x^4 - 15x^2 - 50 = 0$$

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$
$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

Question 9

8. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \qquad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0$$

(2)

8. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \qquad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ

(5)

T04

Inequalities

Questions drawn from printed Question 6 onward, following the paper's progression.

4	32	2	3
questions	marks	families	variants

T04

4 questions and 32 marks occur in the Expertise positions for this topic. 3 of 6 observed Variants are represented.

COVERAGE MAP

Topic 04 Expertise Coverage

Inequalities

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • One-variable Inequalities

V01 Factorable Quadratic Sign Inequality	not selected in this set
V02 Rational Inequality with Critical Points	not selected in this set
V03 Discriminant Condition to a Parameter Range	1 selected

F02 • Graphical Inequality Regions

V01 Shade a Given System of Inequalities	not selected in this set
V02 Write a System from a Shaded Region	1 selected
V03 Derive Boundary Constants, then Write the Region	2 selected

F01

One-variable Inequalities

Reduce an inequality to factors or a quotient, identify all critical values and use sign or graph logic to state the complete solution set.

VARIANT ROUTE

V03

Discriminant Condition to a Parameter Range

1 selected question

1 Expertise questions

6 total marks

FAMILY

Question 10

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C.

(6)

F02

Graphical Inequality Regions

Translate between systems of linear/quadratic inequalities and their coordinate-plane regions, including deriving boundary constants or intersections when necessary.

VARIANT ROUTE

V02 **Write a System from a Shaded Region** — 1 selected question

V03 **Derive Boundary Constants, then Write the Region** — 2 selected questions

3 Expertise questions 26 total marks

FAMILY

Question 11

6.

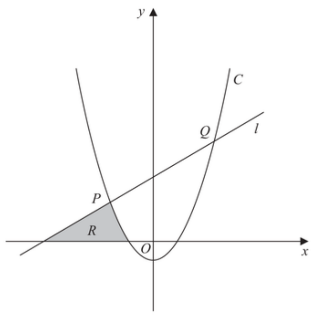


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q, as shown in Figure 3.

(a) Find, using algebra, the coordinates of P and the coordinates of Q.

(4)

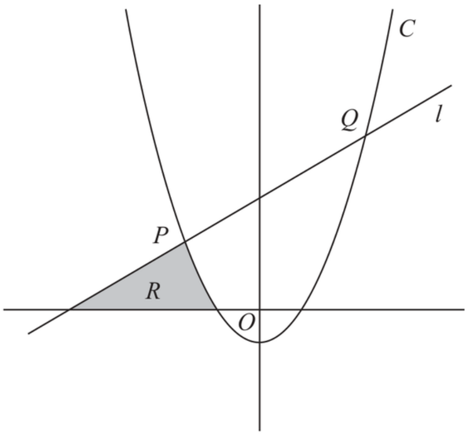


Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q, as shown in Figure 3.

The region R, shown shaded in Figure 3, is bounded by C, l and the x-axis.

(b) Use inequalities to define the region R.

(3)

Question 12

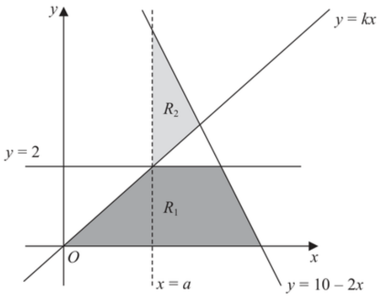


Figure 2

The region R_1, shown shaded in Figure 2, is defined by the inequalities

$$0 \leqslant y \leqslant 2 \qquad y \leqslant 10 - 2x \qquad y \leqslant kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

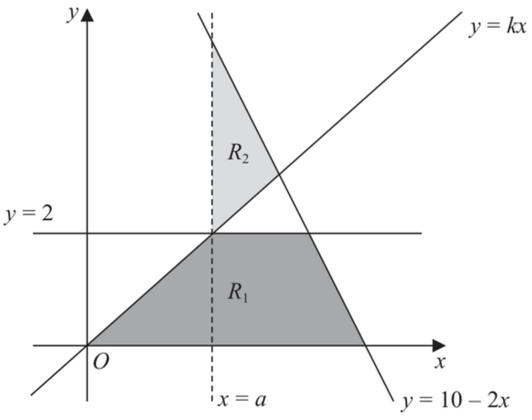


Figure 2

(b) Define the region R_2, also shown shaded in Figure 2, using inequalities.

(2)

9.

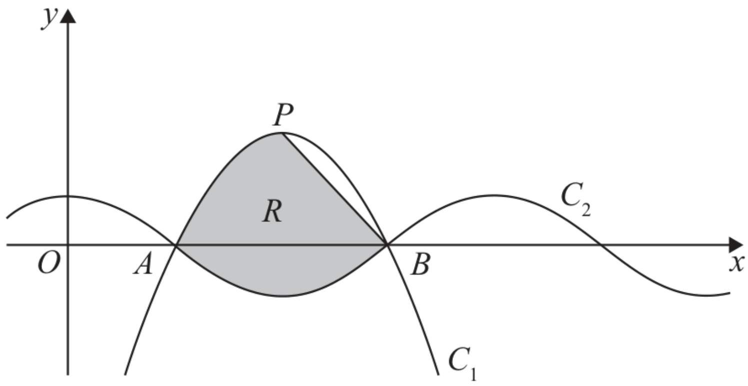


Figure 2

(a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a, b and c are constants to be found.

(3)

9.

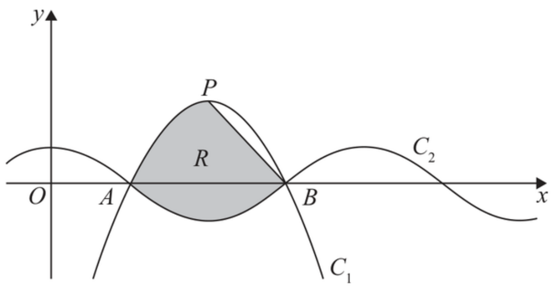


Figure 2

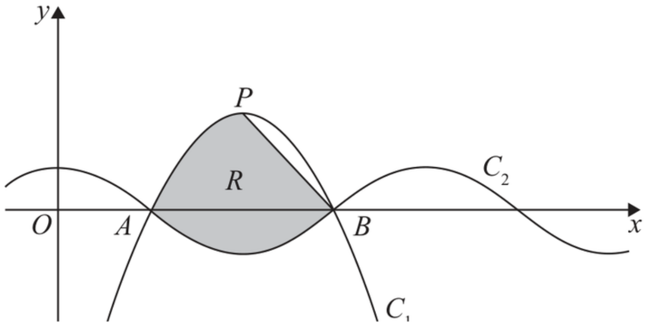
Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

(b) state the coordinates of P

(2)



$$y = 6x - \frac{27}{4} - x^2$$

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

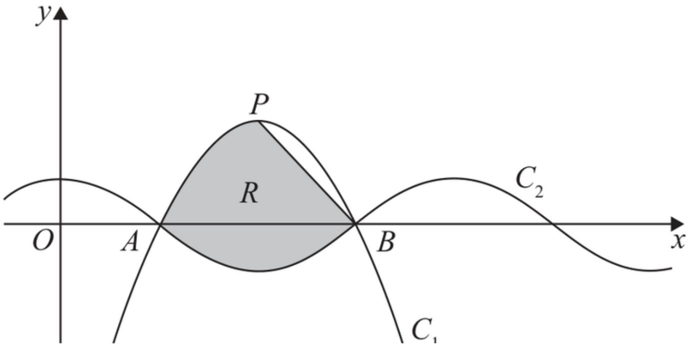
Given that C_1 and C_2 intersect on the x-axis at point A and at point B, as shown in Figure 2,

(c) (i) state the x coordinate of B

(ii) state the value of k

(iii) state the period of C_2

(3)



$$y = 6x - \frac{27}{4} - x^2$$

$$y = \cos(kx)$$

The line segment L joins P and B.
The region R, shown shaded in Figure 2, is bounded by L, C_1 and C_2

(d) Use inequalities to define R.

(5)

T05

Polynomials

Questions drawn from printed Question 6 onward, following the paper’s progression.

3 questions	25 marks	2 families	4 variants

T05

3 questions and 25 marks occur in the Expertise positions for this topic. 4 of 7 observed Variants are represented.

COVERAGE MAP

Topic 05 Expertise Coverage

Polynomials

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Factor and Solve Cubics

V01 Determine a Parameter, then Factor the Cubic	not selected in this set
V02 Common Factor, then Quadratic Factor	not selected in this set

F02 • Build a Polynomial from Roots and Graph Behaviour

V01 Distinct Roots and a Point	1 selected
V02 Repeated Root from Touch Behaviour	2 selected
V03 Intersection by Preserving Common Factors	2 selected

F03 • Transfer Roots Through Transformations or Substitutions

V01 Input-transformation Root Map	1 selected
V02 Substituted-variable Root Transfer with a Domain Filter	not selected in this set

F02

Build a Polynomial from Roots and Graph Behaviour

Translate x-intercepts and touch/cross behaviour into factors with the correct multiplicities, determine a scale from another point, and exploit factor structure in later intersections.

VARIANT ROUTE

V01 **Distinct Roots and a Point** — 1 selected question

V03 **Intersection by Preserving Common Factors** — 2 selected questions

3 Expertise questions — 25 total marks

Question 14

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

The point P is the minimum point of C_1

(b) Deduce the coordinates of P.

(1)

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A, B, C and D are constants.

Given that C_2

- passes through P
- intersects the x-axis at –4, –2 and 3

(c) find, making your method clear, the values of A, B, C and D.

(5)

6.

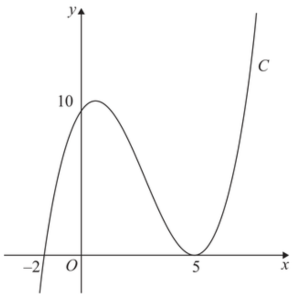


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a cubic function in x.

The curve C

- cuts the x-axis at $(-2, 0)$ and cuts the y-axis at $(0, 10)$
- touches the x-axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $\mathrm{f}\left(\frac{1}{3}x\right) = 0$

(ii) $\mathrm{f}(x - 3) = 0$

(2)

Figure 3 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a cubic function in x.

The curve C

- cuts the x-axis at $(-2, 0)$ and cuts the y-axis at $(0, 10)$
- touches the x-axis at $(5, 0)$

as shown in Figure 3.

(b) Find an expression for $\mathrm{f}(x)$. You should leave your answer in factorised form.

(3)

Figure 3 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a cubic function in x.

The curve C

- cuts the x-axis at $(-2, 0)$ and cuts the y-axis at $(0, 10)$
- touches the x-axis at $(5, 0)$

as shown in Figure 3.

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(*Solutions based entirely on calculator technology are not acceptable.*)

(4)

Question 16

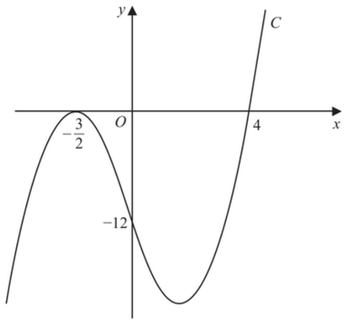


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = f(x)$.

The curve C

- touches the x-axis at $-\frac{3}{2}$
- cuts the x-axis at 4
- cuts the y-axis at -12

Given that $f(x)$ is a cubic function,

(a) find $f(x)$. You may leave the answer in a factorised form.

(3)

The curve C

- touches the x-axis at $-\frac{3}{2}$
- cuts the x-axis at 4
- cuts the y-axis at -12

Given that $f(x)$ is a cubic function,

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q.

(b) Use algebra to find the exact x coordinates of P and Q.
Give each answer in simplest form.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

T06

Graphs of Functions

Questions drawn from printed Question 6 onward, following the paper’s progression.

15	133	2	6
questions	marks	families	variants

T06

15 questions and 133 marks occur in the Expertise positions for this topic. 6 of 8 observed Variants are represented.

COVERAGE MAP

Topic 06 Expertise Coverage

Graphs of Functions

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra	6 selected
V02 Reconstruct a cubic from graph data, then intersect another curve	1 selected
V03 Horizontal-line intersections and geometric separation	1 selected

F02 • Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant	7 selected
V02 Reciprocal-square symmetry and four-intersection parameter range	2 selected
V03 Count real roots by comparing polynomial and reciprocal sketches	1 selected

F03 • Curves, intersections, regions and models

V01 Line-quadratic intersections and region inequalities	not selected in this set
V02 Compare contextual quadratic and linear models with a valid domain	not selected in this set

F01

Polynomial graph grammar

Use factorised polynomial structure, root multiplicity, intercepts, end behaviour and algebraic form to sketch, read or reconstruct cubic graphs.

VARIANT ROUTE

V01 **Factorised cubic: intercepts, multiplicity, sign and linked algebra** — 3 selected questions

V02 **Reconstruct a cubic from graph data, then intersect another curve** — 1 selected question

V03 **Horizontal-line intersections and geometric separation** — 1 selected question

5 Expertise questions — 47 total marks

FAMILY

Question 17

10.

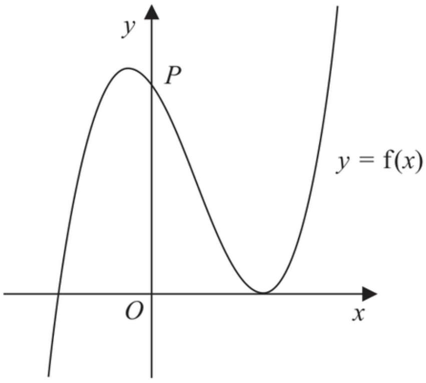


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

(a) Deduce the values of x for which $f(x) \leqslant 0$

(2)

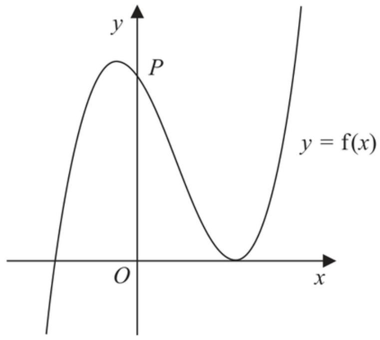


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (2x + 5)(x - 3)^2$$

The curve crosses the y-axis at the point P, as shown.

(b) Expand $\mathrm{f}(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)

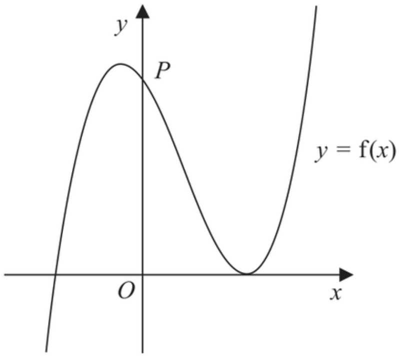


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (2x + 5)(x - 3)^2$$

(c) Hence, or otherwise, find

(i) the coordinates of P,

(ii) the gradient of the curve at P.

(2)

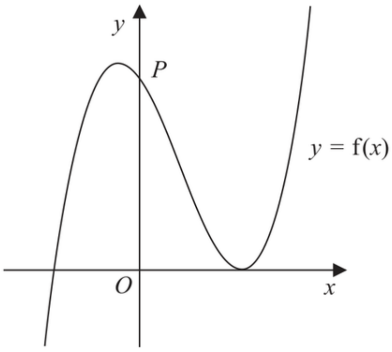


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = \text{f}(x)$, where

$$\text{f}(x) = (2x + 5)(x - 3)^2$$

The curve with equation $y = \text{f}(x)$ is translated two units in the positive x direction to a curve with equation $y = \text{g}(x)$.

(d) (i) Find $\text{g}(x)$, giving your answer in a simplified factorised form.

(ii) Hence state the y intercept of the curve with equation $y = \text{g}(x)$.

(3)

10. The curve C_1 has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (4x - 3)(x - 5)^2$$

(a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

10. The curve C_1 has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (4x - 3)(x - 5)^2$$

(b) Hence or otherwise

(i) find the values of x for which $\mathrm{f}\left(\frac{1}{4}x\right) = 0$

(ii) find the value of the constant p such that the curve with equation $y = \mathrm{f}(x) + p$ passes through the origin.

(2)

10. The curve C_1 has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (4x - 3)(x - 5)^2$$

A second curve C_2 has equation $y = \mathrm{g}(x)$, where $\mathrm{g}(x) = \mathrm{f}(x + 1)$

(c) (i) Find, in simplest form, $\mathrm{g}(x)$. You may leave your answer in a factorised form.

(ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

Question 19

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C.

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2(x + 1)(x - 3)^2$$

(b) Write $\mathrm{f}(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(c) Hence, find the equation of the tangent to C at the point where $x = \dfrac{1}{3}$

(4)

Question 20

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x-axis at the origin and cuts the x-axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

The curve C_2 has equation $y = 1.2x(8 - x)$

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

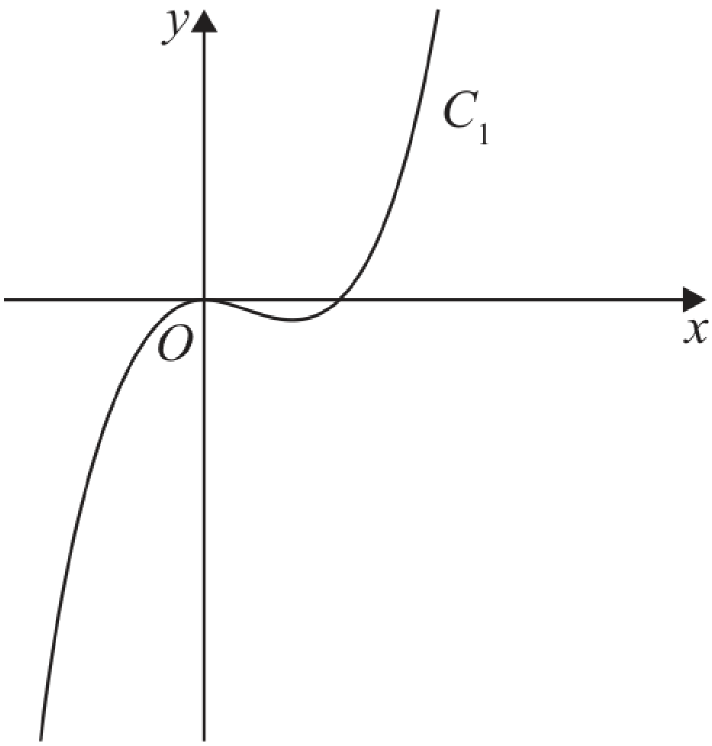


Diagram 1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x-axis at the origin and cuts the x-axis at $x = 4$
- passes through the point $(10, 120)$

The curve C_2 has equation $y = 1.2x(8 - x)$

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect.
Show each stage of your working.

(5)

Question 21

8.

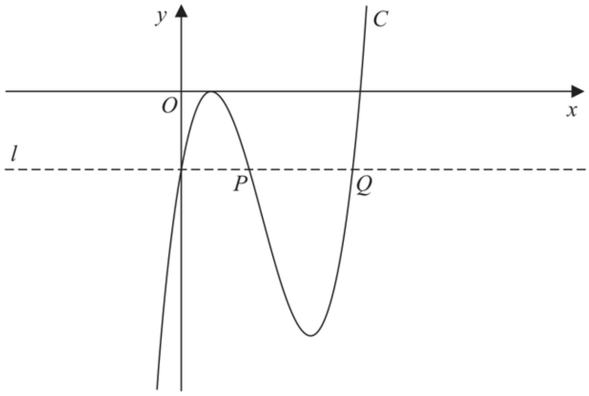


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (3x - 2)^2 (x - 4)$$

(a) Deduce the values of x for which $\mathrm{f}(x) > 0$

(1)

Figure 4 shows a sketch of part of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (3x - 2)^2 (x - 4)$$

(b) Expand $\mathrm{f}(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)

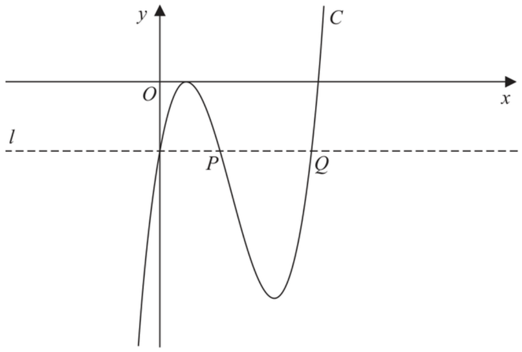


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (3x - 2)^2 (x - 4)$$

The line l, also shown in Figure 4, passes through the y intercept of C and is parallel to the x-axis.

The line l cuts C again at points P and Q, also shown in Figure 4.

(c) Using algebra and showing your working, find the length of line PQ. Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(5)

F02

Reciprocal graphs and intersection counts

Sketch translated reciprocal or reciprocal-square curves and connect their asymptotes, intercepts and symmetry with equations whose number of intersections is controlled algebraically or graphically.

VARIANT ROUTE

V01 **Shifted a/x graph with line-intersection discriminant** — 7 selected questions

V02 **Reciprocal-square symmetry and four-intersection parameter range** — 2 selected questions

V03 **Count real roots by comparing polynomial and reciprocal sketches** — 1 selected question

10 Expertise questions — 86 total marks

Question 22

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

(a) Sketch a graph of C, stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x-axis.

(3)

6. The curve C has equation $y = \dfrac{4}{x} + k$, where k is a positive constant.

The line with equation $y = 10 - 2x$ is a tangent to C.

(b) Find the possible values for k.

(5)

Question 23

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \qquad k > 0 \qquad x \neq 0$$

(2)

(b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \qquad k > 0 \qquad x \neq 0$$

stating the coordinates of the point of intersection with the x-axis and, in terms of k, the equation of the horizontal asymptote.

(3)

(c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \qquad k > 0 \qquad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k.

(5)

Question 25

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2 - x} \qquad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y-axis

(3)

$$y = \frac{1}{2 - x} \qquad x \neq 2$$

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geqslant 0$$

(ii) find the range of possible values for k.

(6)

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

(a) Sketch the graph of C.

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k.

(3)

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

(b) find the range of possible values of k.

(5)

Question 27

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C,

(b) find the range of possible values of k.

(5)

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \qquad x \neq 2k$$

where k is a positive constant.

(a) Sketch a graph of C.
Show on your sketch the coordinates of the point where C crosses a coordinate axis.
(3)

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \qquad x \neq 2k$$

where k is a positive constant.

The line l has equation $x + y = 12$

Given that l is a tangent to C,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

The curve C has equation

$$y = \frac{3k}{x - 2k} \qquad x \neq 2k$$

where k is a positive constant.

The line l has equation $x + y = 12$

$k^2 - 15k + 36 = 0$

The line l meets C at a point P.

Given that P lies below the x-axis,

(c) find the coordinates of P.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C.

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k.

(5)

Question 30

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

(a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

The curve C_2 has equation $y = \dfrac{A}{x}$ where A is a constant.

(b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

(c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \dfrac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k, where appropriate.

(5)

Given that k is a positive constant such that $0 < k < 4$

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \dfrac{k}{x^2}$

(b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

T07

Transformations of Functions

Questions drawn from printed Question 6 onward, following the paper's progression.

5	43	3	5
questions	marks	families	variants

T07

5 questions and 43 marks occur in the Expertise positions for this topic. 5 of 6 observed Variants are represented.

COVERAGE MAP

Topic 07 Expertise Coverage

Transformations of Functions

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Coordinate maps under function notation

V01 Translations and reflections of key points and asymptotes	3 selected
V02 Horizontal input scale/reflection with equations and inequalities	3 selected
V03 Combined input transformation and intercept/multiplicity mapping	not selected in this set

F02 • Transformed algebraic graphs and exact intersections

V01 Square-root transformations with exact surd intersection	2 selected
V02 Translated reciprocal graph followed by line intersection	1 selected

F03 • Transformations on trigonometric graphs

V01 Identify a sine graph and transform a labelled extremum	1 selected

F01

Coordinate maps under function notation

Transform an already-known graph and carry its labelled features through the coordinate map.

VARIANT ROUTE

V02

Horizontal input scale/reflection with equations and inequalities

1 selected question

1 Expertise questions 8 total marks

FAMILY

7.

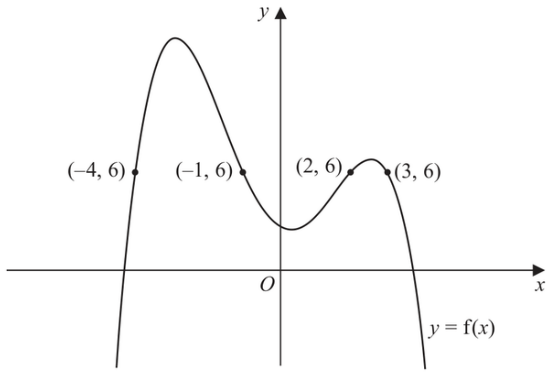


Figure 1

Figure 1 shows the curve with equation $y = \mathrm{f}(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$\mathrm{f}(x) < 6$$

(3)

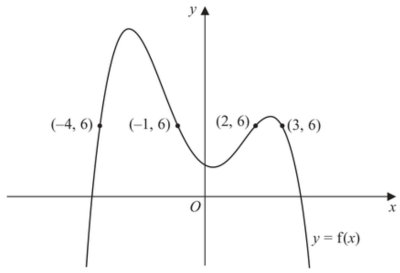


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(b) State the largest solution of the equation

$$f(2x) = 6$$

(1)

..

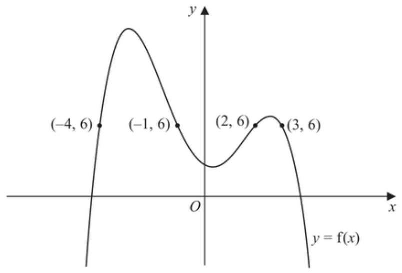


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P, Q, R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geqslant 6 \quad \text{and} \quad x < 0$$

(4)

F02

Transformed algebraic graphs and exact intersections

Describe the graph transformations of a named algebraic function and then solve exactly where two transformed versions meet.

VARIANT ROUTE

V01

Square-root transformations with exact surd intersection — 2 selected questions

V02

Translated reciprocal graph followed by line intersection — 1 selected question

3 Expertise questions — 29 total marks

FAMILY

Question 33

$$f(x) = \sqrt{x} \qquad\qquad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

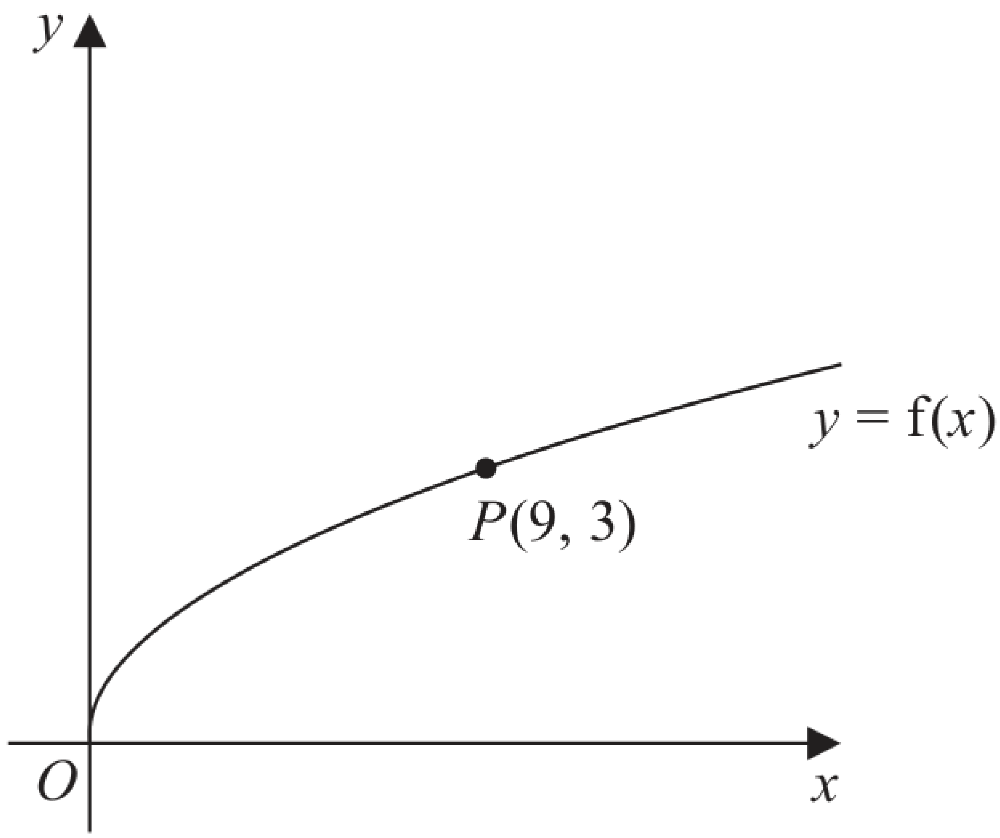


Diagram 1

$\mathrm{f}(x) = \sqrt{x} \qquad x > 0$

The graph of $y = \mathrm{f}(2x)$ meets the graph of $y = \mathrm{f}(x) + 3$ at the point Q.

(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3\left(\sqrt{2} + 1\right)$$

(3)

$\mathrm{f}(x) = \sqrt{x} \qquad\qquad x > 0$

The graph of $y = \mathrm{f}(2x)$ meets the graph of $y = \mathrm{f}(x) + 3$ at the point Q.

$\sqrt{x} = 3\left(\sqrt{2} + 1\right)$

(c) Hence find, in simplest form, the coordinates of Q.

(3)

Question 34

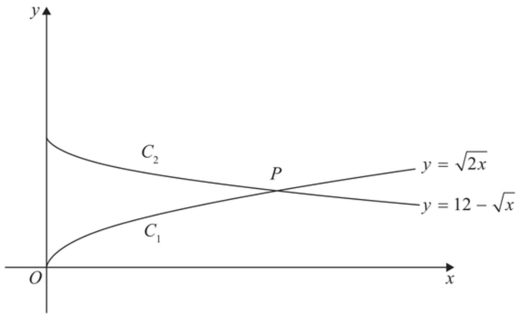


Figure 4

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

(i) the graph with equation $y = \sqrt{x}$ onto C_1

(ii) the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

The graphs C_1 and C_2 meet at the point P, as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12\left(\sqrt{2} - 1\right)$$

(ii) Hence find, in simplest form, the exact coordinates of P.

(6)

7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \qquad x \neq 0$$

(2)

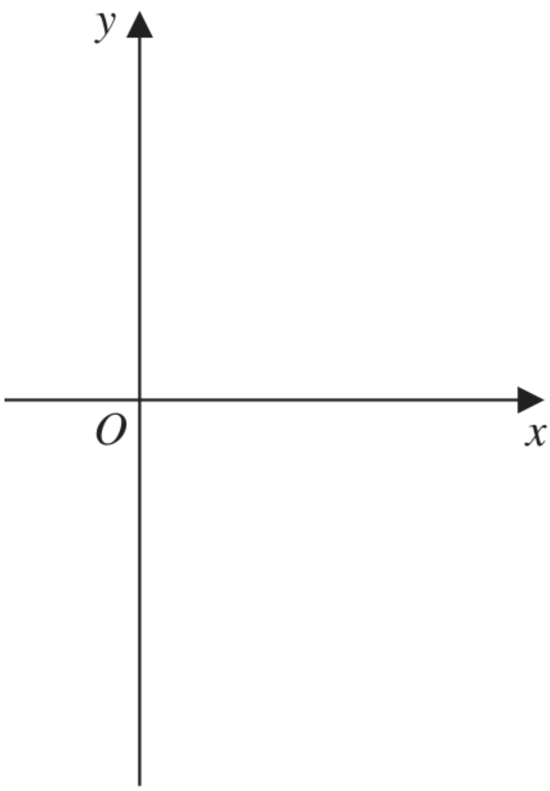


Diagram 1

$$y = \frac{6}{x} \qquad x \neq 0$$

The curve C is transformed onto the curve with equation $y = \dfrac{6}{x-2} \qquad x \neq 2$

(b) Fully describe this transformation.

(2)

The curve with equation

$$y = \frac{6}{x-2} \qquad x \neq 2$$

and the line with equation

$$y = kx + 7 \qquad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q.

Given that the x coordinate of point P is -4

(c) find the value of k,

(2)

The curve with equation

$$y = \frac{6}{x - 2} \qquad x \neq 2$$

and the line with equation

$$y = kx + 7 \qquad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q.

Given that the x coordinate of point P is -4

(d) find, using algebra, the coordinates of point Q.

(*Solutions relying entirely on calculator technology are not acceptable*.)

(4)

Transformations on trigonometric graphs

Recognise a basic sinusoidal graph and map a labelled point under phase translation and output reflection/scaling.

VARIANT ROUTE

V01

Identify a sine graph and transform a labelled extremum

1 selected question

1 Expertise questions 6 total marks

FAMILY

Question 36

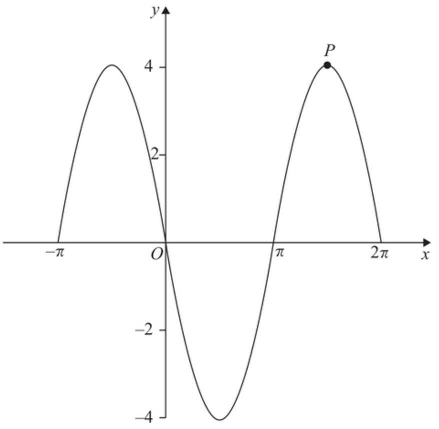


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

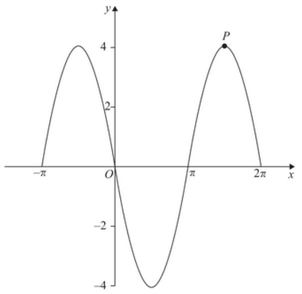


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

The point P lies on $y = f(x)$ and is shown in Figure 3.

(b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \dfrac{\pi}{6}\right)$

(2)

(ii) $y = -\dfrac{1}{2}f(x)$

(2)

T08

Straight Lines and Coordinate Geometry

Questions drawn from printed Question 6 onward, following the paper's progression.

5	46	3	7
questions	marks	families	variants

T08

5 questions and 46 marks occur in the Expertise positions for this topic. 7 of 11 observed Variants are represented.

COVERAGE MAP

Topic 08 Expertise Coverage

Straight Lines and Coordinate Geometry

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Linear models

V01 Fit a two-point linear model and interpret or predict	not selected in this set

F02 • Constructing related lines

V01 Line through two points in a requested form	1 selected
V02 Parallel or perpendicular line through a point	3 selected
V03 Perpendicular bisector from endpoint coordinates	not selected in this set
V04 Exact line work with surd coordinates	not selected in this set

F03 • Coordinate intersections, shapes, distance and area

V01 Related-line intersection and perpendicular distance	1 selected
V02 Coordinate polygons from vectors and diagonal midpoints	2 selected
V03 Coordinate lengths and areas under geometric constraints	2 selected

F04 • Straight lines with curves and regions

V01 Build line/curve boundaries and describe a region	not selected in this set
V02 Tangent parallel to a constructed line	1 selected
V03 Parameterized line-quadratic intersections with a perpendicular constraint	1 selected

F03

Coordinate intersections, shapes, distance and area

Use line equations together with midpoint, vector, distance and area facts to solve coordinate-geometry configurations.

VARIANT ROUTE

V02 **Coordinate polygons from vectors and diagonal midpoints** — 2 selected questions

V03 **Coordinate lengths and areas under geometric constraints** — 1 selected question

3 Expertise questions — 26 total marks

Question 37

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x-axis at $R(8,0)$ and is parallel to l_1

(a) Find the equation of l_2, writing your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x-axis at $R(8,0)$ and is parallel to l_1

The line l_1 cuts the x-axis at P and the y-axis at Q.

Given that $PQRS$ is a parallelogram, find

(b) the area of $PQRS$,

(3)

The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x-axis at $R(8,0)$ and is parallel to l_1

The line l_1 cuts the x-axis at P and the y-axis at Q.

Given that $PQRS$ is a parallelogram, find

(c) the coordinates of S.

(2)

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

(a) Find the gradient of l_1

(1)

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

(b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

The lines l_1 and l_2 intersect at the point M.

(c) Using algebra and showing all your working, find the coordinates of M.

(*Solutions relying on calculator technology are not acceptable*.)

(3)

The line l_1 has equation

$$2x - 5y + 7 = 0$$

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

Given that the diagonals of a square $ABCD$ meet at M,

(d) find the coordinates of the point C.

(2)

9. Given that

- the point A has coordinates (4, 2)
- the point B has coordinates (15, 7)
- the line l_1 passes through A and B

(a) find an equation for l_1, giving your answer in the form $px + qy + r = 0$ where p, q and r are integers to be found.

(3)

- the point A has coordinates (4, 2)
- the point B has coordinates (15, 7)

The line l_2 passes through A and is parallel to the x-axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C.

(4)

- the point A has coordinates (4, 2)
- the point B has coordinates (15, 7)

The line l_2 passes through A and is parallel to the x-axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(c) Hence find the minimum possible area of triangle ABC.

(2)

Straight lines with curves and regions

Combine line construction with quadratic equations, tangent gradients or graph inequalities.

VARIANT ROUTE

V02 **Tangent parallel to a constructed line** — 1 selected question

V03 **Parameterized line-quadratic intersections with a perpendicular constraint** — 1 selected question

2 Expertise questions — 20 total marks

8.

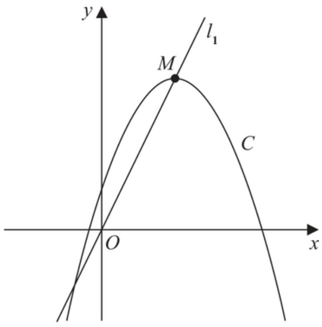


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C.

(a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a, b and c are constants to be found.

(ii) Hence, or otherwise, state the coordinates of M.

(5)

8.

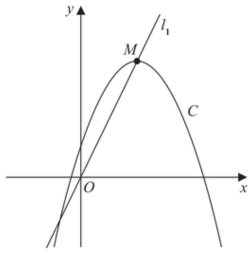


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C.

The line l_1 passes through O and M, as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

(b) Find an equation for l_2

(5)

Question 41

10.

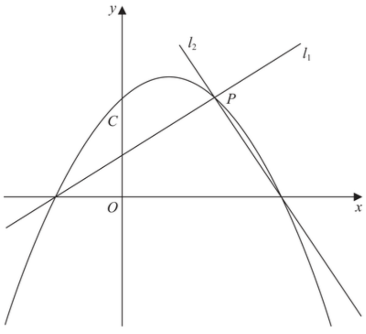


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \qquad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x-axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b, the coordinates of P

(3)

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x + 2)(x - b) \qquad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x-axis,

(b) find, in terms of b, an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

$y = -\frac{1}{4}(x + 2)(x - b)$ where b is a positive constant

- has gradient $\frac{1}{2}$

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x-axis,

$y = -2x + \frac{5b}{2} - 4$

(d) Hence, or otherwise, find the value of b

(2)

T10

Radians

Questions drawn from printed Question 6 onward, following the paper's progression.

11	108	3	10
questions	marks	families	variants

T10

11 questions and 108 marks occur in the Expertise positions for this topic. 10 of 12 observed Variants are represented.

COVERAGE MAP

Topic 10 Expertise Coverage

Radians

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Sector measures and inverse sector problems

V01 Direct arc, area and perimeter of one sector	5 selected
V02 Recover radius or angle from arc length or sector area	3 selected
V03 Concentric, annular and linked sectors	3 selected
V04 Coupled Sector Area-and-Perimeter Constraints	1 selected

F02 • Recovering the central angle and choosing the correct arc

V01 Angle from straight lines, symmetry or rotational partition	2 selected
V02 Angle from triangle trigonometry	3 selected
V03 Major, minor and reflex sector selection	5 selected

F03 • Composite sector designs

V01 Minor sector joined to one or more polygons	4 selected
V02 Major sector with an attached or removed polygon	3 selected
V03 Repeated rotational sector design	1 selected

F04 • Circular segments and chords

V01 Segment area	not selected in this set
V02 Segment perimeter and chord	not selected in this set

F01

Sector measures and inverse sector problems

Apply s = r theta and A = 1/2 r^2 theta directly or inversely, including linked and concentric sectors.

VARIANT ROUTE

V03 **Concentric, annular and linked sectors** — 3 selected questions

V04 **Coupled Sector Area-and-Perimeter Constraints** — 1 selected question

4 Expertise questions — 32 total marks

Question 42

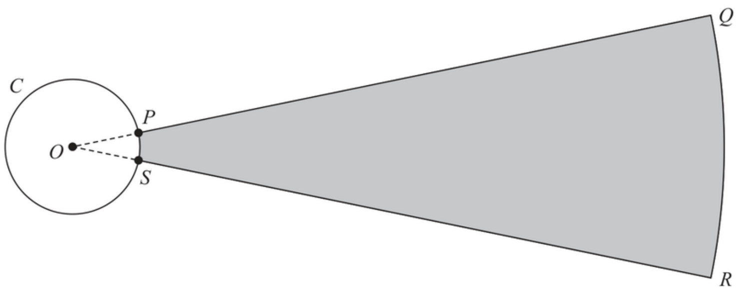


Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

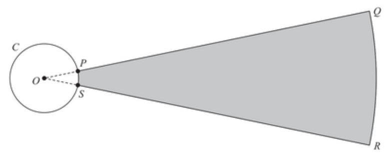
- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2\text{ m}$

(1)

9.

Diagram **NOT** accurately drav



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Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

Given also that

- the target area, $PQRS$, is $90\,\text{m}^2$
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

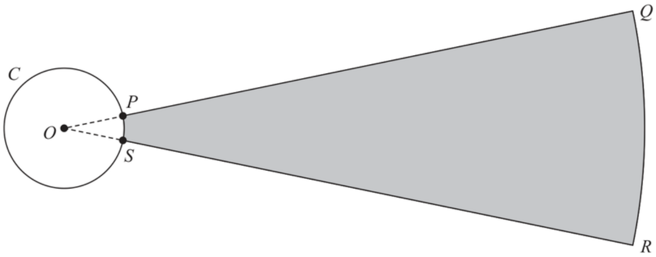


Figure 3

P and S are points on C

$OPQRSO$ is a sector of a circle with centre O

the length of arc PS is 0.72 m

the size of angle POS is 0.6 radians

$OP = 1.2$ m

length $PQ = x$ metres

$5x^2 + 12x - 1500 = 0$

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

Question 43

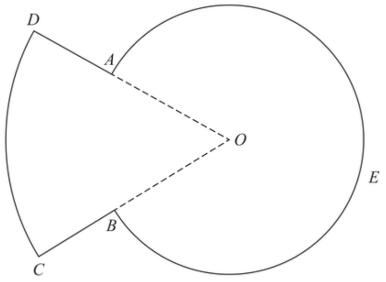


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

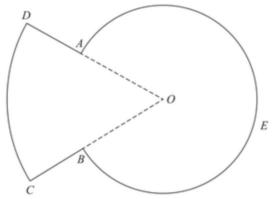
The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O.

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)



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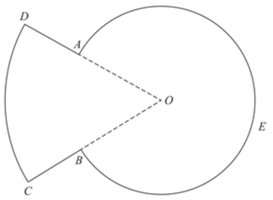
Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA : AO = 3 : 5$

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)



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Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9\,\text{m}$
- $DA : AO = 3 : 5$

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

Question 44

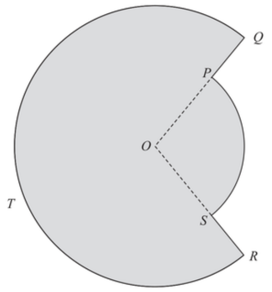


Figure 2

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector *POS* of a circle centre *O* joined to a major sector *QORTQ* of a different circle, also with centre *O*.

Given that

- angle *POS* is 1.65 radians
- the area of sector *POS* is $30\,\text{m}^2$
- $PQ = 2.8\,\text{m}$

(a) show that, to 3 significant figures, $OQ = 8.83\,\text{m}$

(3)

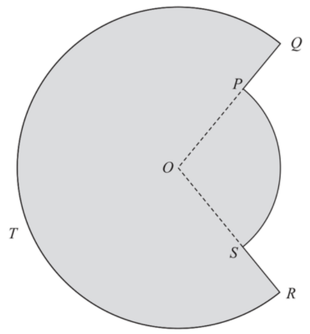


Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O.

Given that

- angle POS is 1.65 radians
- the area of sector POS is $30\,\text{m}^2$
- $PQ = 2.8\,\text{m}$

$OQ = 8.83\,\text{m}$

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

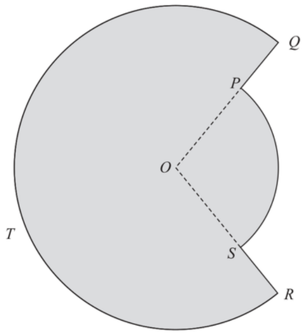


Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector *POS* of a circle centre *O* joined to a major sector *QORTQ* of a different circle, also with centre *O*.

Given that

- angle *POS* is 1.65 radians
- the area of sector *POS* is $30\,\text{m}^2$
- $PQ = 2.8\,\text{m}$

$OQ = 8.83\,\text{m}$

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

Question 45

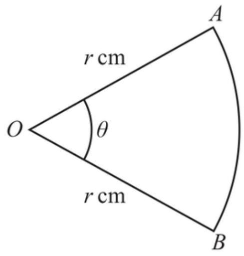


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sector AOB of a circle, centre O.

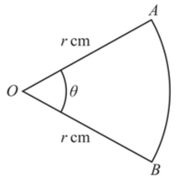
The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(a) write down two different equations involving r and θ.

(2)



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The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

Given that $r > 2$

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ.
Write each answer in the form $a + b\sqrt{c}$ where a, b and c are integers.

(*Solutions relying on calculator technology are not acceptable.*)

(5)

Recovering the central angle and choosing the correct arc

Obtain the radian angle from a diagram before applying sector formulae, and decide whether the required region uses theta or 2 pi minus theta.

VARIANT ROUTE

V02

Angle from triangle trigonometry

1 selected question

1 Expertise questions 10 total marks

7.

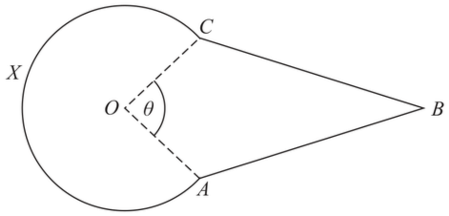


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O.

In the design

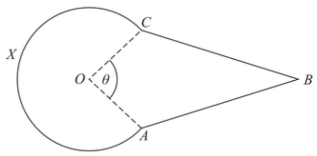
- $OA = OC = 0.6\,\text{m}$
- $AB = CB = 1.4\,\text{m}$
- Angle OAB = Angle OCB = 2 radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(a) show that $\theta = 1.64$ radians to 3 significant figures,

(4)

7.



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In the design

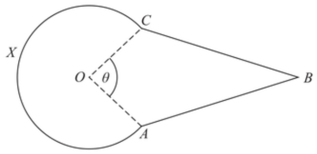
- $OA = OC = 0.6\,\text{m}$
- $AB = CB = 1.4\,\text{m}$
- Angle OAB = Angle OCB = 2 radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(b) find the perimeter of the sign, in metres to 2 significant figures,

(2)

7.



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In the design

- $OA = OC = 0.6\,\text{m}$
- $AB = CB = 1.4\,\text{m}$
- Angle OAB = Angle OCB = 2 radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(c) find the area of the sign, in m^2 to 2 significant figures.

(4)

F03

Composite sector designs

Assemble or subtract sector, triangle, trapezium, rectangle and radial-edge contributions to find a complete area or perimeter.

VARIANT ROUTE

V01 **Minor sector joined to one or more polygons** — 3 selected questions

V02 **Major sector with an attached or removed polygon** — 2 selected questions

V03 **Repeated rotational sector design** — 1 selected question

6 Expertise questions — 66 total marks

FAMILY

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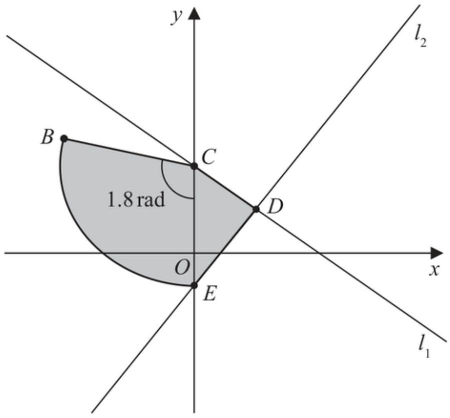


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y-axis at the point C, as shown in Figure 3.

(a) State the y coordinate of C.

(1)

7.

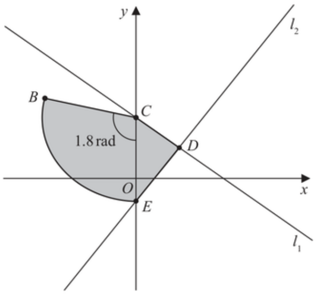


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y-axis at the point C, as shown in Figure 3.

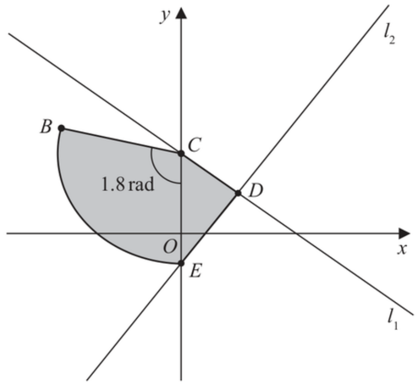
The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y-axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\dfrac{14}{3}$

(3)



The line l_1 has equation $4y + 3x = 48$

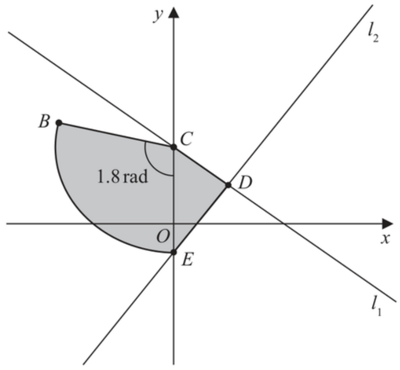
the y coordinate of E is $-\dfrac{14}{3}$

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE.

(3)



The line l_1 has equation $4y + 3x = 48$

The point $D(8, 6)$ lies on l_1

the y coordinate of E is $-\dfrac{14}{3}$

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE.

(d) Calculate the exact area of the region $CBED$.

(3)

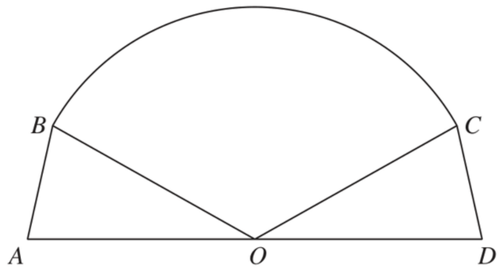


Figure 1

Figure 1 shows the plan view for the design of a stage.

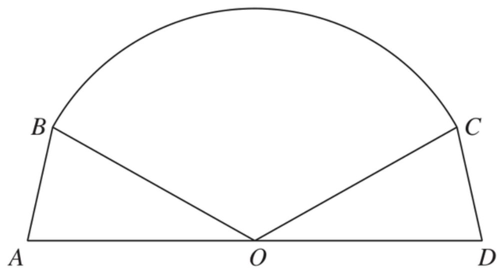
The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m, to 2 decimal places,

(2)



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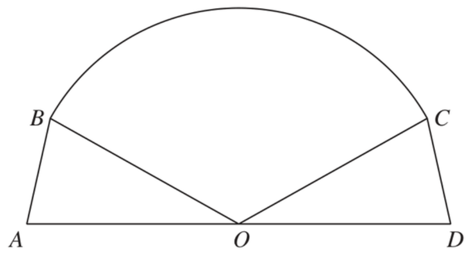
The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length $12.5\,\text{m}$

(b) find the size of angle AOB, in radians, to 2 decimal places.

(1)



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The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

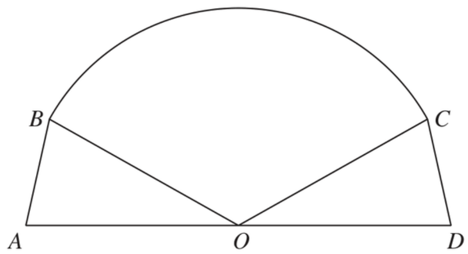
Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length $12.5\,\text{m}$

Hence find

(c) the total area of the stage, giving your answer, in m^2, to one decimal place,

(3)



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The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length $12.5\,\text{m}$

Hence find

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)

o.

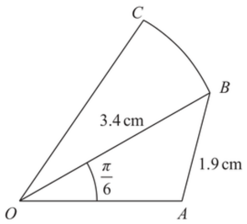


Figure 1

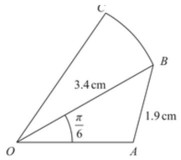
Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4\,\text{cm}$
- $AB = 1.9\,\text{cm}$
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB, giving your answer in radians to 4 significant figures,

(3)



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The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(b) find the area of triangle OAB, in cm^2, giving your answer to 3 significant figures.

(2)

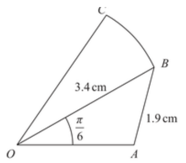


Figure 1

Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle *OAB* joined to a sector *OBC* of a circle with centre *O*
In the design

- $OB = 3.4\text{ cm}$
- $AB = 1.9\text{ cm}$
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

Given that the ratio of the area of sector *OBC* to the area of triangle *OAB* is $3:2$

(c) show that angle *BOC* is 0.462 radians to 3 significant figures.

(3)

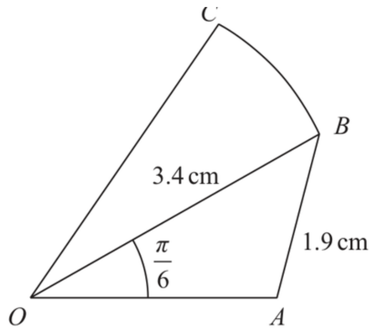


Figure 1

Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

angle $OAB > \dfrac{\pi}{2}$ radians

angle BOC is 0.462 radians to 3 significant figures.

(d) Hence find the perimeter of the badge, in cm, to the nearest integer.

(5)

8.

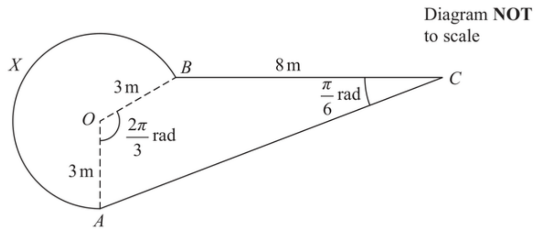


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8\text{ m}$
- $OA = OB = 3\text{ m}$
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

8.

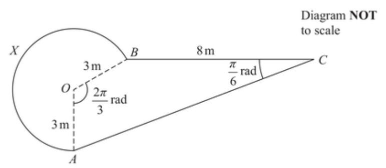


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(b) Calculate the exact area of the triangle AOB.

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

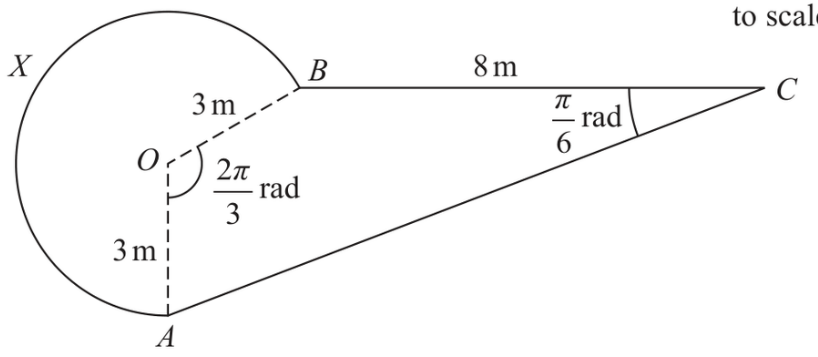


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector *AOBX* of a circle centre *O* joined to a quadrilateral *AOBC*.

length *AB* is $3\sqrt{3}$ m.

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

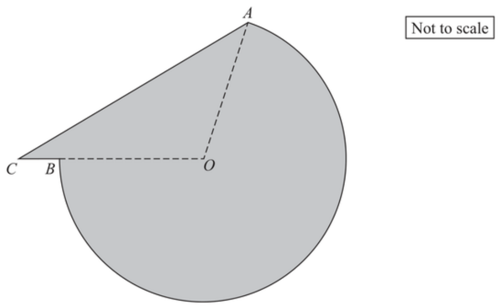


Figure 2

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

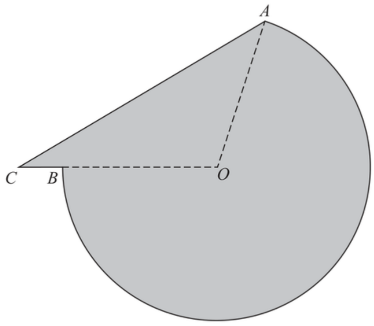
The area consists of the major sector AOB of a circle centre O joined to a triangle AOC.

Given that

- $AO = OB = 15\,\text{m}$
- $BC = 2\,\text{m}$
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)



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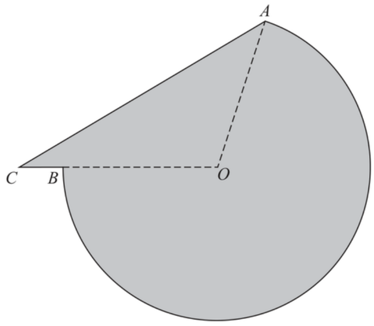
Given that

- $AO = OB = 15\,\text{m}$
- $BC = 2\,\text{m}$
- CBO is a straight line
- angle $ACO = 0.6$ radians

angle COA is 1.847 radians to 3 decimal places.

(b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)



The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector *AOB* of a circle centre *O* joined to a triangle *AOC*.

Given that

- $AO = OB = 15\,\text{m}$
- $BC = 2\,\text{m}$
- *CBO* is a straight line
- angle $ACO = 0.6$ radians

angle *COA* is 1.847 radians to 3 decimal places.

(c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)

Question 52

8.

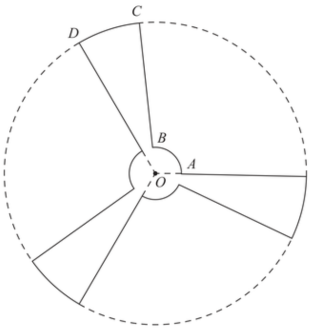


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

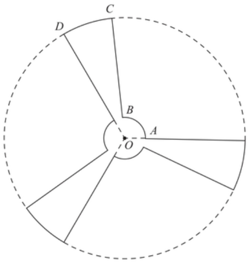
The fan consists of three identical sections congruent to *OABCDO*, shown in Figure 3, where

- *OABO* is a sector of a circle with centre *O* and radius 9 cm
- *OBCDO* is a sector of a circle with centre *O* and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc *AB* is 15 cm,

(a) show that the length of the arc *CD* is 35.9 cm to one decimal place.

(3)



The fan consists of three identical sections congruent to $OABCDO$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \dfrac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

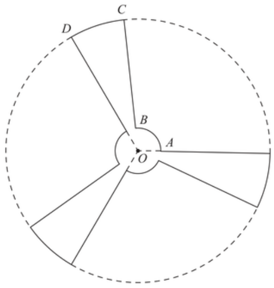
arc CD is 35.9 cm to one decimal place.

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)



The fan consists of three identical sections congruent to $OABCDO$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \dfrac{2\pi}{3}$ radians

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

T11

Trigonometric Functions

Questions drawn from printed Question 6 onward, following the paper's progression.

13	81	4	10
questions	marks	families	variants

T11

13 questions and 81 marks occur in the Expertise positions for this topic. 10 of 11 observed Variants are represented.

COVERAGE MAP

Topic 11 Expertise Coverage

Trigonometric Functions

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Reading and identifying trigonometric graphs

V01 Key points, range and period	8 selected
V02 Infer amplitude, frequency or equation from features	5 selected
V03 Key points under vertical, horizontal and reflection transformations	7 selected

F02 • Symmetry, reference angles and symbolic solutions

V01 Reference-angle and sign identities	2 selected
V02 Solve a trigonometric equation in terms of alpha	2 selected
V03 Link sine and cosine solutions by phase	1 selected

F03 • Counting roots and graphical intersections

V01 Count solutions of a level equation over long intervals	1 selected
V02 Count intersections with a line or non-trigonometric curve	4 selected
V03 Parameter values giving an exact root count	not selected in this set

F04 • Constructing trigonometric sketches

V01 Sketch transformed sine and cosine	3 selected
V02 Sketch tangent with vertical asymptotes	3 selected

F01

Reading and identifying trigonometric graphs

Use amplitude, period, sign, phase and key points to read a graph or infer its equation and parameters.

VARIANT ROUTE

V02 **Infer amplitude, frequency or equation from features** — 1 selected question

V03 **Key points under vertical, horizontal and reflection transformations** — 7 selected questions

8 Expertise questions — 49 total marks

Question 53

.

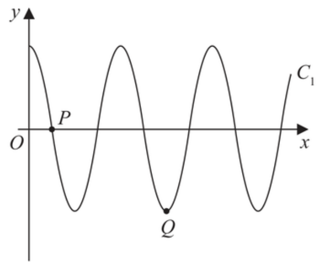


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3\cos\left(\frac{x}{n}\right)^{\circ} \qquad x \geqslant 0$$

where n is a constant.

The curve C_1 cuts the positive x-axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

10.

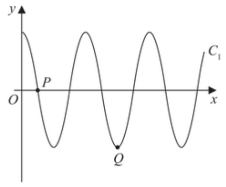


Figure 4

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$$y = 3\cos\left(\frac{x}{n}\right)^{\circ} \qquad x \geqslant 0$$

where n is a constant.

The curve C_1 cuts the positive x-axis for the first time at point $P(270, 0)$, as shown in Figure 4.

The point Q, shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q.

(2)

The curve C_2 has equation $y = 2\sin x° + k$, where k is a constant.

The point $R\left(a, \dfrac{12}{5}\right)$ and the point $S\left(-a, -\dfrac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k.

(2)

Question 54

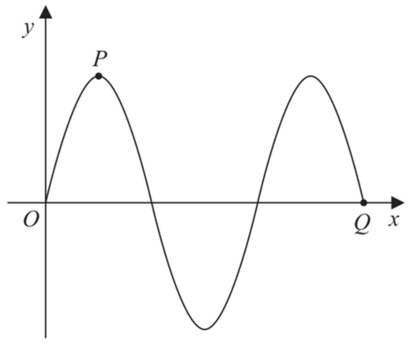


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3\sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

(i) the coordinates of P,

(ii) the coordinates of Q.

(3)

A different curve C_2 has equation $y = 3\sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R.

(2)

Question 55

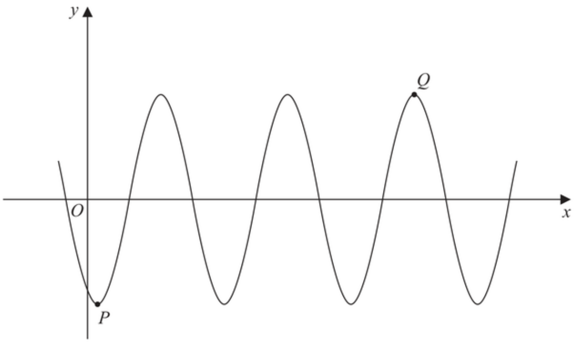


Figure 4

Figure 4 shows part of the curve with equation

$$y = A\cos(x - 30)°$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

(a) Write down the value of A.

(1)

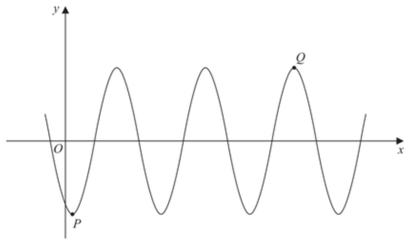


Figure 4

Figure 4 shows part of the curve with equation

$$y = A\cos(x - 30)°$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates $(30, -3)$ as shown in Figure 4.

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q.

(3)

9. (i)

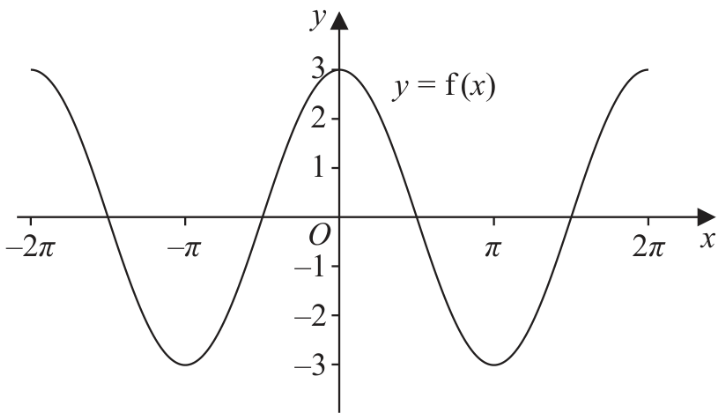


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

9. (i)

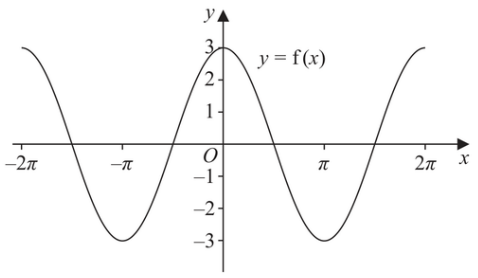


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

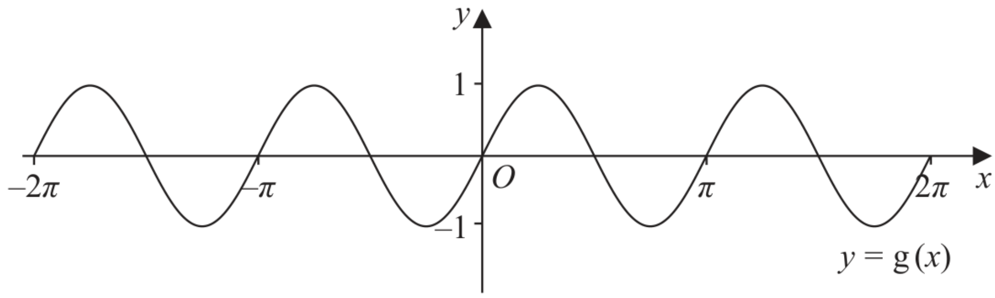


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = \mathrm{g}(x)$

(a) Write down an expression for $\mathrm{g}(x)$

(2)

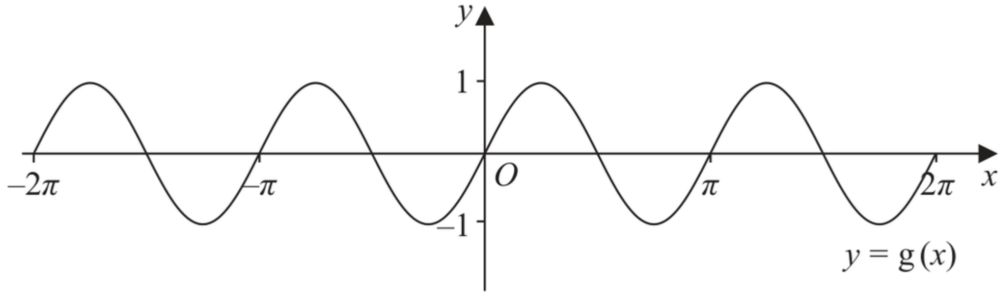


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = \mathrm{g}(x)$

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = \mathrm{g}(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

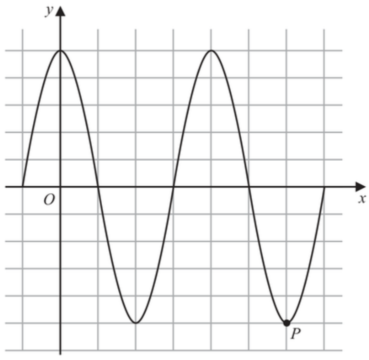


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5\cos x$$

with x being measured in degrees.

The point P, shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5\cos x - 2$

(ii) C_2 has equation $y = -5\cos x$

(4)

11.

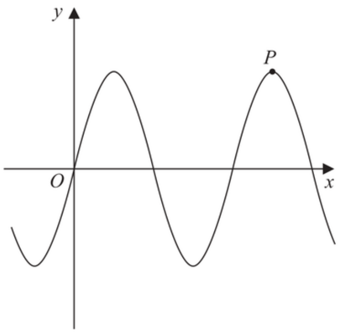


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12\sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P.

(2)

The curve C_2 has equation

$$y = 12\sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12\sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A,

(ii) the smallest possible value of B.

(2)

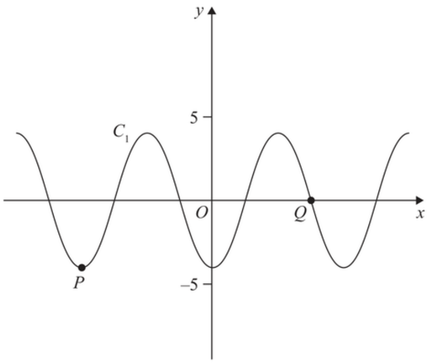


Figure 3

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4\cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

(i) the coordinates of P

(ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4\cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

(ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4\cos x = 5 - \frac{10}{\pi}x$$

giving a reason for your answer.

(2)

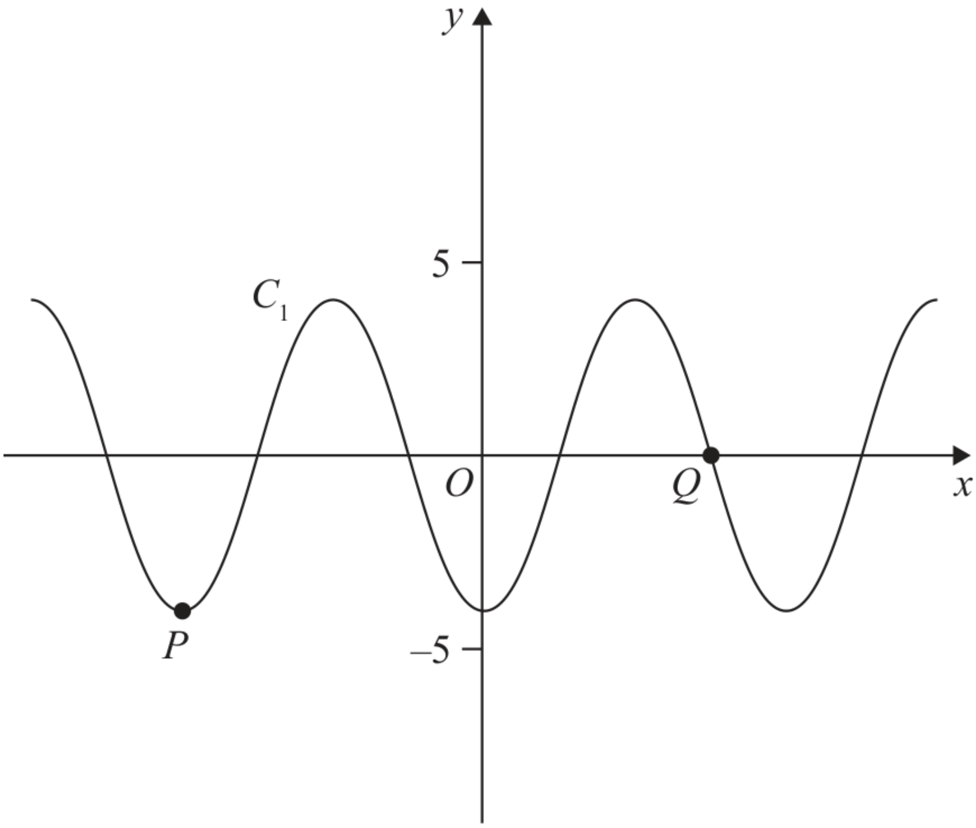


Diagram 1

Question 60

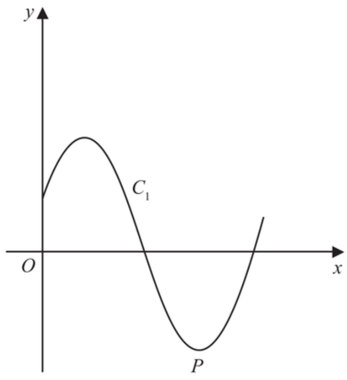


Figure 4

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leqslant x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(a) State the x coordinate of P.

(1)

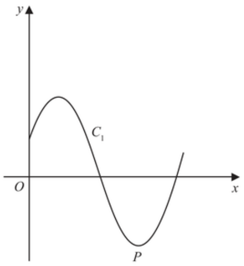


Figure 4

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leqslant x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(b) State the number of roots of the equation

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leqslant x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leqslant x < 2\pi$

(c) State the coordinates of Q.

(2)

F02

Symmetry, reference angles and symbolic solutions

Use trigonometric symmetry and phase links to express exact values or solutions in terms of a given angle alpha.

VARIANT ROUTE

V02 **Solve a trigonometric equation in terms of alpha** — 1 selected question

V03 **Link sine and cosine solutions by phase** — 1 selected question

2 Expertise questions — 12 total marks

FAMILY

Question 61

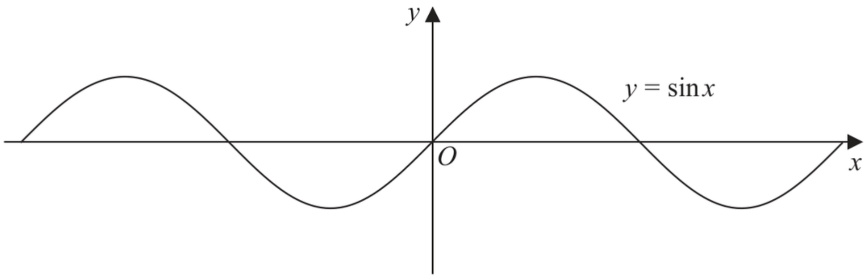


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin \alpha = p$, where $0 < \alpha < 90°$

(a) state, in terms of p, the value of

(i) $2\sin(180° - \alpha)$

(ii) $\sin(\alpha - 180°)$

(iii) $3 + \sin(180° + \alpha)$

(3)

Given that $\sin\alpha = p$, where $0 < \alpha < 90°$

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

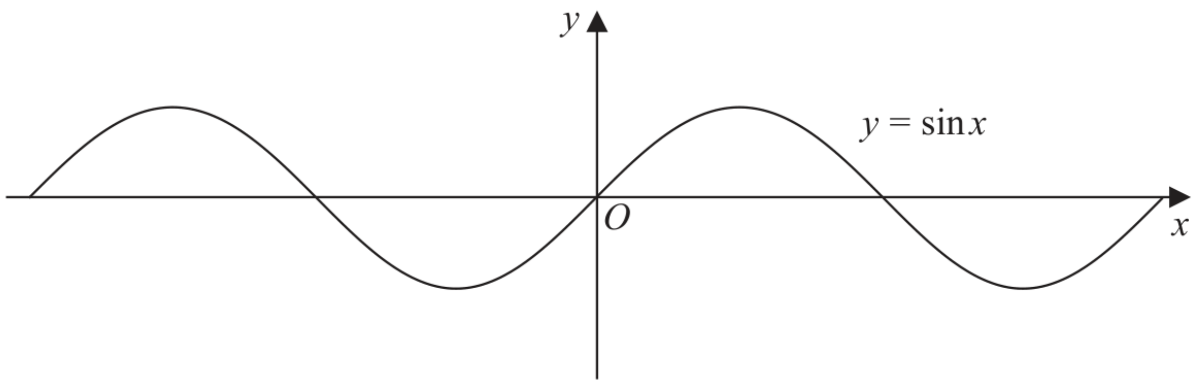


Diagram 1

Given that $\sin\alpha = p$, where $0 < \alpha < 90°$

(c) Hence find, in terms of α, the x coordinates of any points in the interval $0 < x < 180°$ where

$$\sin 2x = p$$

(3)

Question 62

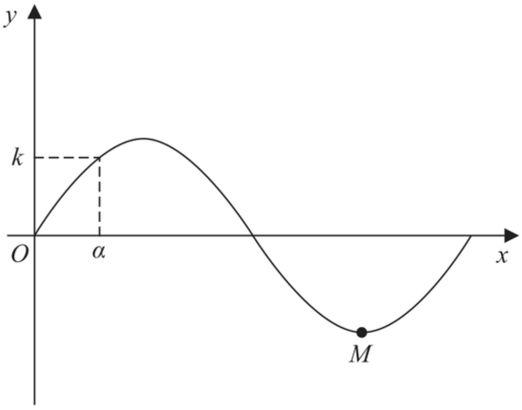


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C.

(a) State the period of C.

(1)

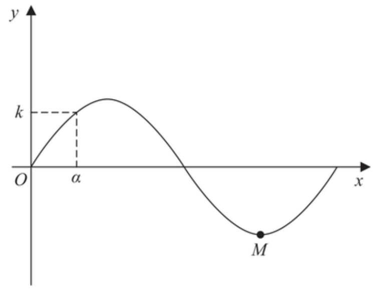


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C.

(b) State the coordinates of M.

(1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α.

Find, in terms of α,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,

(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

(2)

F03

F03

Counting roots and graphical intersections

Count solutions from periodic graphs, endpoint rules and intersections with horizontal lines or non-trigonometric curves.

VARIANT ROUTE

V02

Count intersections with a line or non-trigonometric curve

3 selected questions

3 Expertise questions 20 total marks

FAMILY

Question 63

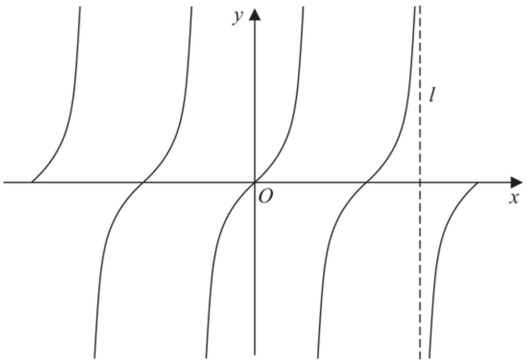


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

The line l, shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l.

(1)

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \qquad -2\pi \leqslant x \leqslant 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leqslant x \leqslant 2\pi$

(4)

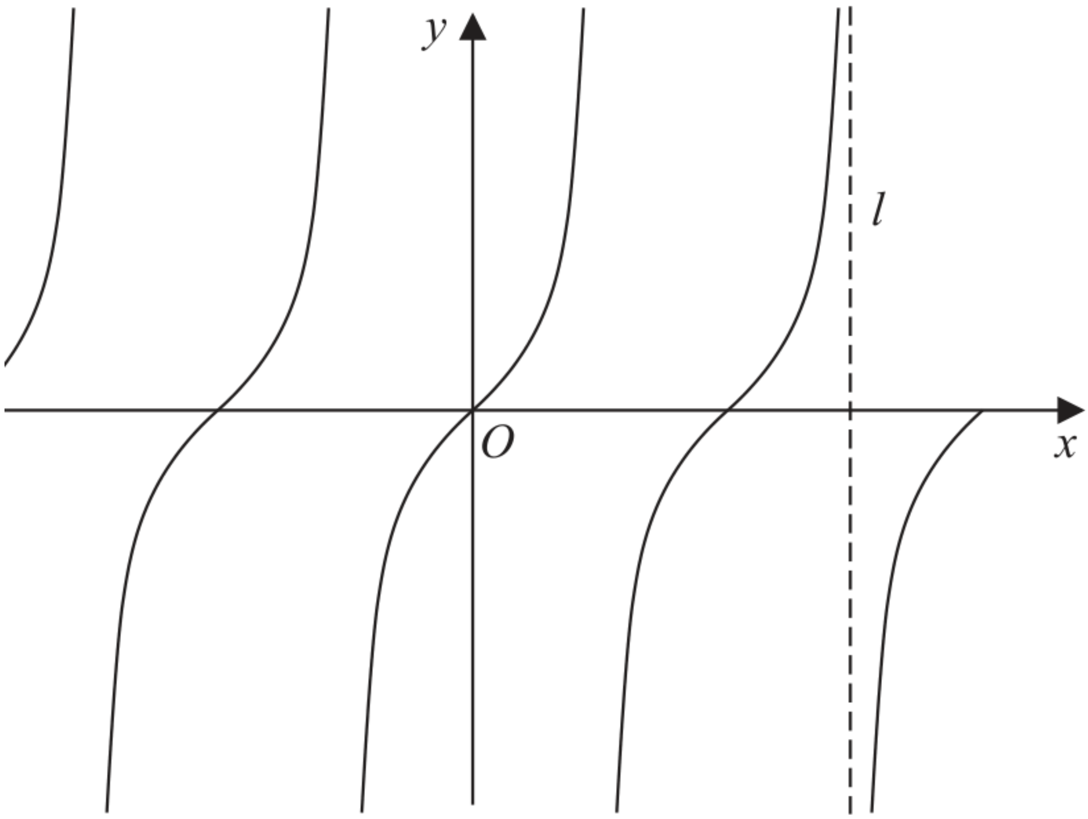


Diagram 1

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leqslant x \leqslant 40\pi$

(ii) $-10\pi \leqslant x \leqslant \frac{5}{2}\pi$

(2)

Question 64

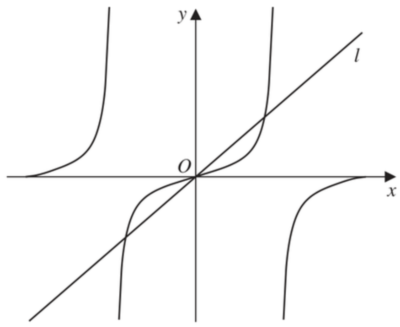


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

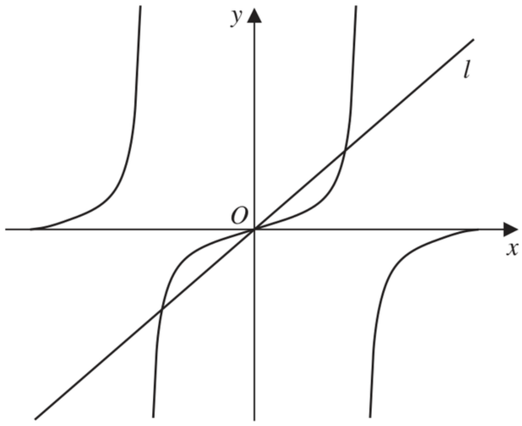


Figure 3

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

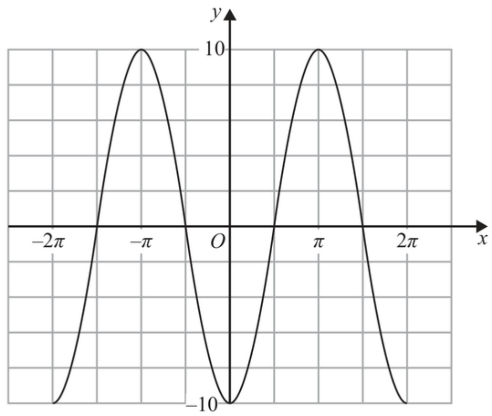


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = \mathrm{f}(x)$.

(a) Write down an expression for $\mathrm{f}(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

(3)

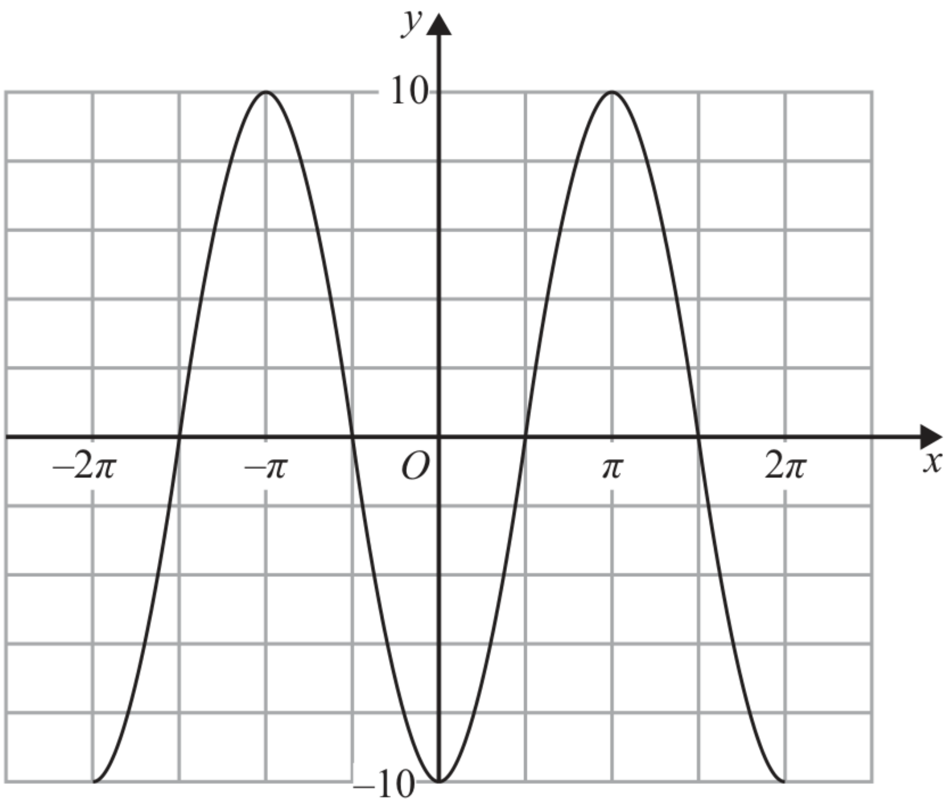


Diagram 1

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$\mathrm{f}(x) = \tan x \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

giving a reason for your answer.

(4)

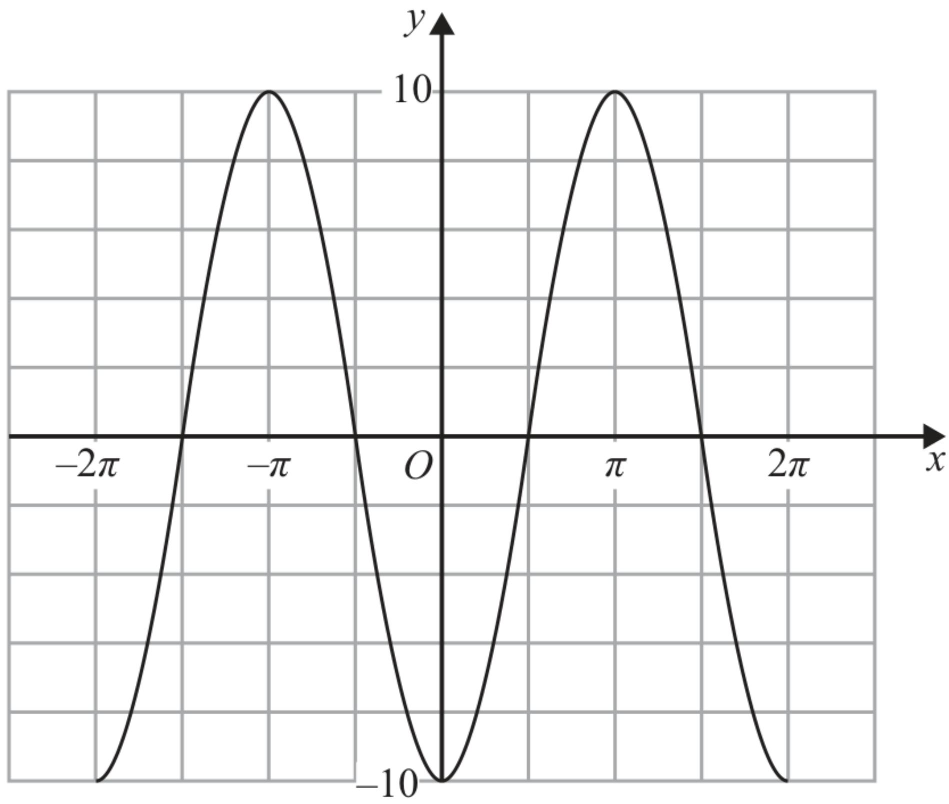


Diagram 2

EXPERTISE ROUTE • TOPIC 12

T12

Differentiation

Questions drawn from printed Question 6 onward, following the paper's progression.

11	106	5	10
questions	marks	families	variants

T12

11 questions and 106 marks occur in the Expertise positions for this topic. 10 of 10 observed Variants are represented.

COVERAGE MAP

Topic 12 Expertise Coverage

Differentiation

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Power-rule differentiation and algebraic preparation

V01 Rewrite then find the first derivative — 3 selected

V02 Second derivative and higher-derivative comparison — 2 selected

F02 • Tangents, normals and linked line geometry

V01 Tangent or normal at a given point — 5 selected

V02 Find the contact point from a gradient or line — 1 selected

V03 Linked normals, tangents and parallel-gradient points — 3 selected

F03 • Equations, inequalities and extrema involving derivatives

V01 Stationary-point equation — 1 selected

V02 Solve a prescribed first- or second-derivative value — 3 selected

V03 Gradient inequality or extremum of the derivative — 2 selected

F04 • Chord gradients and the derivative limit

V01 Chord gradient tending to the tangent gradient — 1 selected

F05 • Global curve synthesis with tangents

V01 Tangent re-intersects the original curve — 2 selected

F02

Tangents, normals and linked line geometry

Use derivative gradients with point-gradient form, reciprocal gradients and linked tangent or normal conditions.

VARIANT ROUTE

V02 **Find the contact point from a gradient or line** — 1 selected question

V03 **Linked normals, tangents and parallel-gradient points** — 3 selected questions

4 Expertise questions — 46 total marks

FAMILY

Question 66

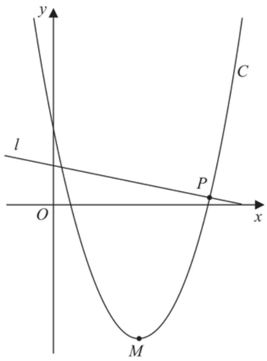


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

(a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

$$y = \frac{1}{2}x^2 - 10x + 22$$

The point M is the minimum turning point of C, as shown in Figure 3.

(b) Deduce the coordinates of M

(2)

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

The line l is the normal to C at the point P, as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

(c) (i) find the coordinates of P

(ii) find the value of k

(6)

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

The line l is the normal to C at the point P, as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

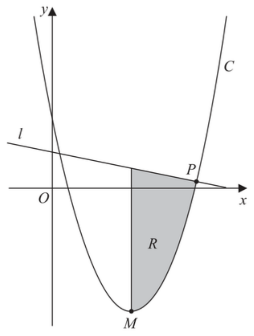


Figure 4

Figure 4 is a copy of Figure 3. The finite region R, shown shaded in Figure 4, is bounded by l, C and the line through M parallel to the y-axis.

(d) Identify the inequalities that define R.

(3)

Question 67

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

(a) Find $\frac{dy}{dx}$

(2)

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

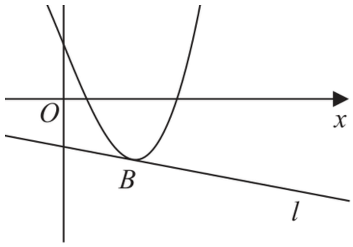
The line l, shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B,

(b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)



$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

The line l, shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B,

$12x^2 + 4x - 33 = 0$

(c) Hence find the x coordinate of B, justifying your answer.

(2)

Given that the y intercept of l is -1

(d) find the value of k.

(4)

Question 68

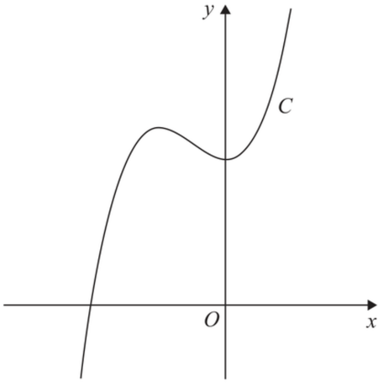


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $\mathrm{f}(x) \geqslant 0$

(1)

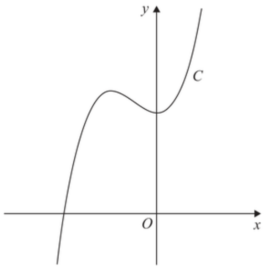


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (x+5)(3x^2 - 4x + 20)$$

(b) Find $\mathrm{f}'(x)$ giving your answer in simplest form.

(3)

$$\text{f}(x) = (x+5)(3x^2 - 4x + 20)$$

The point $R(-4, 84)$ lies on C.

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P.

(4)

$$\mathrm{f}(x) = (x+5)(3x^2 - 4x + 20)$$

The point $R(-4, 84)$ lies on C.

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(d) Find the point to which R is transformed when the curve with equation $y = \mathrm{f}(x)$ is transformed to the curve with equation,

(i) $y = \mathrm{f}(x-3)$

(ii) $y = 4\mathrm{f}(x)$

(2)

Question 69

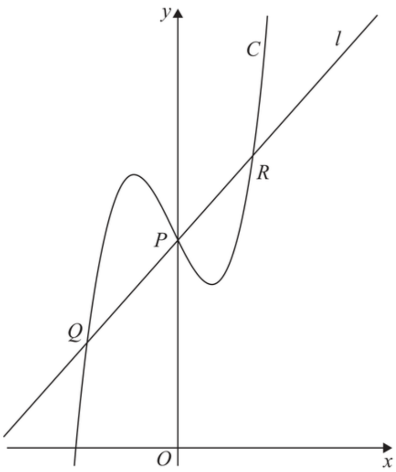


Figure 4

(a) Find, using algebra, the x coordinate of point Q.

(6)

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q.

(6)

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

The point T lies on C.

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T.

(4)

F03

Equations, inequalities and extrema involving derivatives

Solve conditions imposed on f prime or f double prime and interpret domain, sign or minimum-gradient information.

VARIANT ROUTE

V01 **Stationary-point equation** — 1 selected question

V02 **Solve a prescribed first- or second-derivative value** — 2 selected questions

V03 **Gradient inequality or extremum of the derivative** — 2 selected questions

5 Expertise questions 44 total marks

FAMILY

9. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \qquad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

Question 71

7. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \qquad x > 0$$

(a) Solve the equation

$$f(x) = 9$$

(4)

7. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \qquad x > 0$$

(b) Solve the equation

$$f''(x) = 6$$

(5)

7. The curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $\mathrm{f}'(x)$

(ii) $\mathrm{f}''(x)$

(3)

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B.

Given that the x coordinate of A is 5

(b) find the value of k.

(2)

(c) Hence find the coordinates of B.

(3)

Question 73

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

(a) find the value of the constant p.

(2)

$f(x) = (x + 4)(x - 2)(2x - 9)$

Given that the curve with equation $y = f(x + q)$ passes through the origin,

(b) write down the possible values of the constant q.

(2)

$f(x) = (x + 4)(x - 2)(2x - 9)$

(c) Find $f'(x)$.

(4)

(d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18

(3)

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{\mathrm{d}y}{\mathrm{d}x}$ can be written in the form

$$p(x+q)^2 + r$$

where p, q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{\mathrm{d}y}{\mathrm{d}x}$

(ii) the value of x at which this minimum value occurs.

(2)

The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

Given that S is the point on C at which the gradient is a minimum,

(c) find the equation of the tangent to C at S, giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

F04

Chord gradients and the derivative limit

Calculate a chord gradient using an increment h and show that it approaches the tangent gradient as h tends to zero.

VARIANT ROUTE

V01

Chord gradient tending to the tangent gradient

1 selected question

1 Expertise questions 6 total marks

FAMILY

Question 75

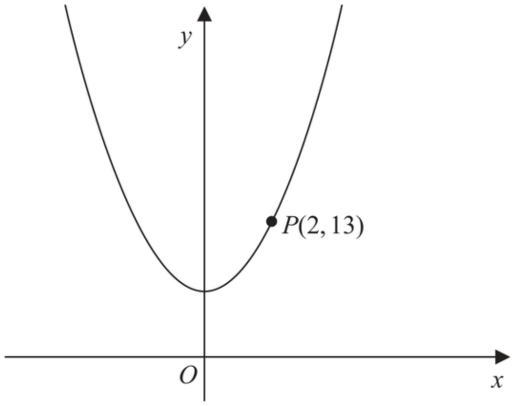


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2,13)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P.

(2)

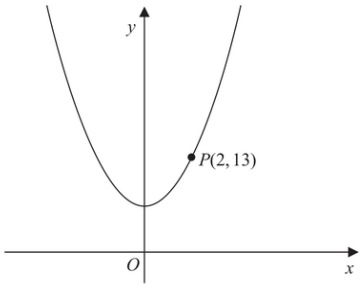


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2,13)$ lies on the curve.

The point Q with x coordinate $2 + h$ also lies on the curve.

(b) Find, in terms of h, the gradient of the line PQ. Give your answer in simplest form.

(3)

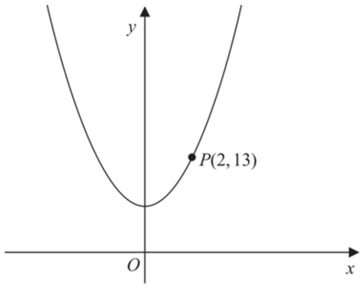


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2,13)$ lies on the curve.

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

F05

Global curve synthesis with tangents

Combine factorised or expanded curve structure with a tangent equation to locate another intersection or interpret the whole curve.

VARIANT ROUTE

V01

Tangent re-intersects the original curve

1 selected question

1 Expertise questions 10 total marks

FAMILY

Question 76

CLASSIFIED SEQUENCE

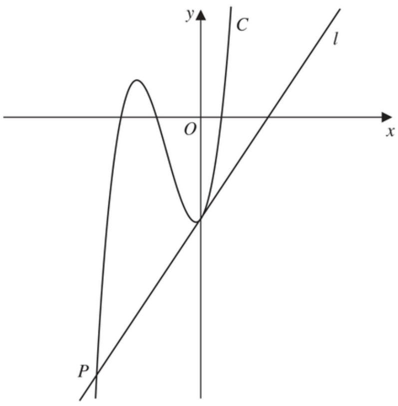


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

(a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

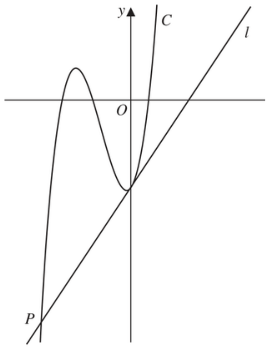


Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

The straight line l is the tangent to C at the point where C cuts the y-axis.

Given that l cuts C at the point P, as shown in Figure 4,

(c) find, using algebra, the x coordinate of P

(*Solutions based on calculator technology are not acceptable*.)

(5)

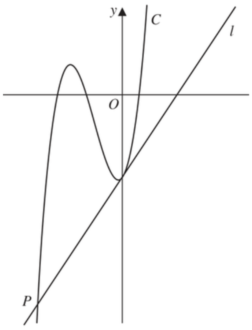


Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

The straight line l is the tangent to C at the point where C cuts the y-axis.

Given that l cuts C at the point P, as shown in Figure 4,

(c) find, using algebra, the x coordinate of P

(*Solutions based on calculator technology are not acceptable*.)

(5)

T13

Integration

Questions drawn from printed Question 6 onward, following the paper's progression.

18	172	4	8
questions	marks	families	variants

T13

18 questions and 172 marks occur in the Expertise positions for this topic. 8 of 10 observed Variants are represented.

COVERAGE MAP

Topic 13 Expertise Coverage

Integration

Each count is the number of selected questions that assess the Variant. A question is printed once and may contribute to more than one Variant route.

F01 • Algebraic preprocessing and indefinite integration

V01 Direct sum of positive, negative and fractional powers	not selected in this set
V02 Expand a product then integrate	not selected in this set
V03 Expand or divide a radical or rational quotient	3 selected

F02 • Recovering a curve from its first derivative

V01 Integrate f prime and use one point	15 selected
V02 Find parameter from gradient or line, then recover the curve	5 selected
V03 Recover the curve and construct a tangent or normal	12 selected

F03 • Recovering from second-derivative information

V01 Integrate twice using gradient and point conditions	3 selected
V02 Use a second-derivative condition to find a parameter	3 selected

F04 • Multi-constraint integration synthesis

V01 Determine multiple coefficients before recovering f	2 selected
V02 Recovery plus a transformation or extra derivative demand	2 selected

F02

Recovering a curve from its first derivative

Integrate f prime, determine the constant from a point and connect the derivative to tangent or normal information when present.

VARIANT ROUTE

V02 **Find parameter from gradient or line, then recover the curve** — 5 selected questions

V03 **Recover the curve and construct a tangent or normal** — 5 selected questions

10 Expertise questions — 98 total marks

FAMILY

8. A curve C has equation $y = \mathrm{f}(x)$.

The point P with x coordinate 3 lies on C

Given

- $\mathrm{f}'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\dfrac{1}{24}x + 5$

(a) show that $k = -5$

(3)

A curve C has equation $y = f(x)$.

The point P with x coordinate 3 lies on C

Given

- $f'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

$k = -5$

(b) Hence find $f(x)$.

(4)

Question 78

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = \mathrm{f}(x),\ x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $\mathrm{f}'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \dfrac{1}{4}$

(ii) find the value of c

(4)

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant

$k = \dfrac{1}{4}$

(b) Hence find the value of $f''(x)$ at P.

(3)

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant

$k = \dfrac{1}{4}$

(c) Find $f(x)$.

(4)

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = f(x), x > 0$

Given that

- $f'(x) = 2x + \dfrac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k.

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = \mathrm{f}(x), x > 0$

Given that

- $\mathrm{f}'(x) = 2x + \dfrac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(c) Find $\mathrm{f}(x)$, writing your answer in simplest form.

(4)

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$, $x > 0$

A point P lies on C.

Given that

- $\mathrm{f}'(x) = 4x^2 + \dfrac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P,

(ii) find, using algebra, the other possible x coordinate of P.

(6)

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x), \quad x > 0$

A point P lies on C.

Given that

- $\mathrm{f}'(x) = 4x^2 + \dfrac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P.

(2)

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$, $x > 0$

A point P lies on C.

Given that

- $\mathrm{f}'(x) = 4x^2 + \dfrac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(c) Find $\mathrm{f}(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

8. A curve C has equation $y = \mathrm{f}(x),\ x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C.

Given that

- $\mathrm{f}'(x) = 10 + \dfrac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(a) (i) find the value of k

(ii) find the equation of the normal to C at P.
Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

8. A curve C has equation $y = \mathrm{f}(x),\ x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C.

Given that

- $\mathrm{f}'(x) = 10 + \dfrac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(b) Find $\mathrm{f}(x)$.

(5)

6. The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $\mathrm{f}'(x) = \dfrac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(a) find the equation of the tangent to C at P. Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

6. The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $\mathrm{f}'(x) = \dfrac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(b) Find, in simplest form, $\mathrm{f}(x)$.

(5)

6. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

6. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(b) Find $\mathrm{f}(x)$, simplifying your answer.

(7)

Question 84

8. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find $f(x)$ giving the coefficients in simplest form.

(5)

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(b) Find $f(x)$.

(6)

10. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = 6x - \dfrac{(2x-1)(3x+2)}{2\sqrt{x}}$
- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P, giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

10. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = 6x - \dfrac{(2x-1)(3x+2)}{2\sqrt{x}}$
- the point $P(4, 12)$ lies on C

(b) find $\mathrm{f}(x)$, giving each term in simplest form.

(6)

F03

Recovering from second-derivative information

Use f double prime either to reconstruct f through two integrations or to determine a parameter in f prime before integrating once.

VARIANT ROUTE

V01 **Integrate twice using gradient and point conditions** — 3 selected questions

V02 **Use a second-derivative condition to find a parameter** — 3 selected questions

6 Expertise questions — 53 total marks

FAMILY

11. A curve has equation $y = \mathrm{f}(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $\mathrm{f}''(x) = \frac{4}{\sqrt{x}} - 3$
- $\mathrm{f}'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P, writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

11. A curve has equation $y = \mathrm{f}(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $\mathrm{f}''(x) = \frac{4}{\sqrt{x}} - 3$
- $\mathrm{f}'(x) = 5$ at P

find

(b) $\mathrm{f}(x)$.

(8)

Question 88

11. A curve has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $\mathrm{f}'(x) = -4$ at P,

(a) find the equation of the normal at P, writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

11. A curve has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $\mathrm{f}'(x) = -4$ at P,

(b) find $\mathrm{f}(x)$.

(8)

11. A curve C has equation $y = \mathrm{f}(x), \quad x > 0$

Given that

- $\mathrm{f}''(x) = 4x + \dfrac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

11. A curve C has equation $y = \mathrm{f}(x), \quad x > 0$

Given that

- $\mathrm{f}''(x) = 4x + \dfrac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(b) find $\mathrm{f}(x)$, writing your answer in simplest form.

(6)

10. A curve has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- $\mathrm{f}'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $\mathrm{f}''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a.

(3)

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(b) Hence find $f(x)$.

(4)

7. The curve C has equation $y = \mathrm{f}(x),\ \ x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{2}{\sqrt{x}} + \dfrac{A}{x^2} + 3$, where A is a constant
- $\mathrm{f}''(x) = 0$ when $x = 4$

(a) find the value of A.

(4)

7. The curve C has equation $y = \mathrm{f}(x), \ x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{2}{\sqrt{x}} + \dfrac{A}{x^2} + 3$, where A is a constant
- $\mathrm{f}''(x) = 0$ when $x = 4$

Given also that

- $\mathrm{f}(x) = 8\sqrt{3}$, when $x = 12$

(b) find $\mathrm{f}(x)$, giving each term in simplest form.

(5)

10. **In this question you must show all stages of your working.**

The curve C has equation $y = \mathrm{f}(x), x > 0$

Given that

- the point $P\left(2, 8\sqrt{2}\right)$ lies on C
- $\mathrm{f}'(x) = 4\sqrt{x^3} + \dfrac{k}{x^2}$ where k is a constant
- $\mathrm{f}''(x) = 0$ at P

(a) find the exact value of k,

(4)

10. **In this question you must show all stages of your working.**

The curve C has equation $y = \mathrm{f}(x), x > 0$

Given that

- the point $P\left(2, 8\sqrt{2}\right)$ lies on C
- $\mathrm{f}'(x) = 4\sqrt{x^3} + \dfrac{k}{x^2}$ where k is a constant
- $\mathrm{f}''(x) = 0$ at P

(b) find $\mathrm{f}(x)$, giving your answer in simplest form.

(4)

F04

Multi-constraint integration synthesis

Resolve several independent constants or combine integration with an additional transformation or derivative demand.

VARIANT ROUTE

V01 **Determine multiple coefficients before recovering f** — 2 selected questions

2 Expertise questions — 21 total marks

FAMILY

Question 93

9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}}\,\mathrm{d}x \qquad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = \mathrm{f}(x)$.

Given

- $\mathrm{f}'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $\mathrm{f}(x)$.

(6)

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $\mathrm{f}'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24$$

(2)

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = \mathrm{f}(x),\ x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $\mathrm{f}'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

Given also that $a + b = -9$

(b) find, in simplest form, $\mathrm{f}(x)$

(7)

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $\mathrm{f}'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

Curve C is transformed to the curve with equation $y = \mathrm{f}(x - 3)$

Given that point P is transformed to the point Q,

(c) state the coordinates of Q.

(1)